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Thesis

Proximity induced magnetism on topological edge states

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In this thesis proximity induced magnetic effects on topological edge states are studied using a real-space tight-binding model of topological insulators based on the Bernevig-Hughes-Zhang formalism. By coupling an additional magnetic layer, the effects of time-reversal symmetry breaking and possible phase transitions can be investigated. The changes in the magnetic properties are also followed by the mapping of the tight-binding Hamiltonian to an extended Heisenberg spin model utilising the Liechtenstein-Katsnelson-Antropov-Gubanov torque formalism. The edge-effects on the magnetic layer, emerging from the finite size of the investigated system, are also studied in different phases of the coupled topological insulator.

1 Introduction

Two-dimensional topological insulators are materials characterised by an insulating bulk and propagating edge states, the presence of which in a finite material is topologically protected against adiabatic deformation of the Hamiltonian through the bulk-boundary correspondence. Experiments made with HgTe/CdTe quantum wells required an accurate low energy theoretical description for the observed phase transition, therefore, the Berneving-Hughes-Zhang model was created [1]. In the heavy-element heterostructures used in the experiments, spin-orbit coupling leads to an inversion of the s-like and p-like bands, transforming the material from its trivial insulator phase to its topological insulator phase. As long as the time-reversal symmetry ($\mathcal{T}$) of the system is preserved, it is capable of hosting a Kramers pair of counter-propagating edge states with opposite spins. These states are topologically protected against elastic backscattering, and also robust against disorder that preserves $\mathcal{T}$, providing a platform for dissipation-less transport.

However, the introduction of a magnetic field breaks the time-reversal symmetry. The following study will focus on a two-dimensional topological insulator coupled to a localised magnetic layer. The presence of the magnetic layer lifts the Kramers degeneracy of the edge states, driving the system into other phases, e.g. trivial or the Quantum Anomalous Hall (QAH) insulator.

To describe the magnetic properties of this localised layer, the Heisenberg model is utilised. While the isotropic Heisenberg model successfully describes collinear magnetic alignments driven by Coulomb interactions [2, 3], it neglects relativistic effects. Since in the investigated system there is a combination of strong spin-orbit coupling and broken inversion symmetry, it makes extension of the spin Hamiltonian necessary [4]. Consequently, to explore possible complex magnetic textures, relativistic extensions such as the Dzyaloshinskii-Moriya interaction (DMI) must be evaluated (they energeti-

cally favour orthogonal spin alignments contrary to the isotropic Heisenberg interactions, thus leading to two competing processes).

Although the topological phase transition can be described by $k \cdot p$ theory, relying on symmetry considerations and applying perturbation theory, which is conventional in the description of semiconductors, large scale numerical calculations of finite materials give rise to the need for a discrete lattice model. Therefore, a real-space tight-binding formalism is constructed to make the extraction of the magnetic exchange parameters possible. The extraction process utilises real-space Green's functions and discrete lattice grid integrals.

With the help of the previously introduced models and methods, we will be able to create a model of a topological insulator coupled to a magnetic layer. The tuning of system parameters, such as the Dirac mass, the interlayer coupling, and the magnetic exchange field, enables us to study the possible phases of the insulator layer and the effects of these phases on the magnetic properties of the material. The exploration of spatial edge effects, arising from a finite system size, will also be a relevant part of this study.

The thesis consists of three parts. In Section 1, the theoretical background of the tight-binding formalism, the quantum Heisenberg model and its relativistic extension, and the BHZ model are presented. In Section 2, the numerical methods are described, including the construction of the real-space Hamiltonian and the extraction of the magnetic exchange interactions. Finally, Section 3 discusses the spatial distribution and the parametric dependence of the calculated interaction terms in both bulk and strip geometric settings.

2 Theoretical background

2.1 Tight-binding model

The tight-binding formalism is extremely useful for accelerating many-body calculations, especially in systems containing more than a few atoms or molecules [5, 6]. In the tight-binding approximation, all effects of the underlying lattice potential and the interactions among electrons are treated as perturbations to the free atomic orbitals. In this case, the Hamiltonian is

$$\hat{H} = \hat{H}_{at} + \Delta\hat{H}\,, \tag{1}$$

where we expect that the perturbation $\Delta\hat{H}$ tends to zero near the centre of the atomic sites. The free atomic orbitals,

$$\hat{H}_{at}\phi_i(\vec{r}) = \varepsilon_i\phi_i(\vec{r})\,, \tag{2}$$

which are eigenstates of the atomic Hamiltonian, have a small overlap between neighbouring atoms, hence the name "tight-binding." Because these states are not eigenstates of the full crystal Hamiltonian, linear combinations of atomic orbitals are used as an approximation of the actual solution [6]:

$$\psi_j(\vec{r}) = \sum_{\vec{R}_i} a_j(\vec{R}_i)\phi_j(\vec{r} - \vec{R}_i)\,. \tag{3}$$

According to Bloch's theorem, these states can change only by a phase factor under spatial translation:

$$\psi(\vec{r} + \vec{R}_m) = \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}_m}\psi(\vec{r})\,, \tag{4}$$

where $\vec{k}$ is the wave vector. Now, by combining Eq. (3) and Eq. (4), one finds the following relation for the expansion coefficients:

$$a_j(\vec{R}_m) = \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}_m}a_j(\vec{0})\,. \tag{5}$$

The final step in creating the correct wave functions is normalisation, after which the defining relation for the coefficients becomes:

$$a_j^*(\vec{0})a_j(\vec{0}) = \frac{1}{N}\cdot\frac{1}{1+\sum_{\vec{R}_p\neq 0}\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}_p}\alpha_j(\vec{R}_p)}\,, \tag{6}$$

$$\alpha_j(\vec{R}_p) = \int \mathrm{d}^3\vec{r}\,\phi_j^*(\vec{r})\phi_j(\vec{r} + \vec{R}_p)\,, \tag{7}$$

where N is the number of lattice sites and $\alpha_j(\vec{R}_p)$ is the overlap integral. A very frequent assumption is neglecting the overlap integral entirely, thus arriving at:

$$a_j(\vec{0}) \approx \frac{1}{\sqrt{N}}\,, \tag{8}$$

$$\psi_j(\vec{r}) \approx \frac{1}{\sqrt{N}} \sum_{\vec{R}_i} \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}_i} \phi_j(\vec{r}-\vec{R}_i)\,. \tag{9}$$

To calculate the Bloch energies, one simply needs to calculate the expectation value of the Hamiltonian with respect to the corresponding Bloch eigenstates:

$$\varepsilon_m = \int \mathrm{d}^3\vec{r}\, \psi_m^*(\vec{r}) \hat{H} \psi_m(\vec{r})\,. \tag{10}$$

Using Eq. (1) and Eq. (3), one arrives at:

$$\begin{aligned}
\varepsilon_m &= \sum_{\vec{R}_p}\sum_{\vec{R}_q} \int \mathrm{d}^3\vec{r}\, \phi_m^*(\vec{r}-\vec{R}_p) a_m^*(\vec{R}_p) \hat{H}_{at} \phi_m(\vec{r}-\vec{R}_q) a_m(\vec{R}_q) \\
&\quad + \sum_{\vec{R}_p} \int \mathrm{d}^3\vec{r}\, \phi_m^*(\vec{r}-\vec{R}_p) a_m^*(\vec{R}_p) \Delta\hat{H}(\vec{r}) \psi_m(\vec{r}) \\
&= a_m^*(\vec{0}) a_m(\vec{0}) N \int \mathrm{d}^3\vec{r}\, \phi_m^*(\vec{r}) \hat{H}_{at} \phi_m(\vec{r}) \\
&\quad + a_m^*(\vec{0}) \sum_{\vec{R}_p} \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}_p} \int \mathrm{d}^3\vec{r} \phi_m^*(\vec{r}-\vec{R}_p) \Delta\hat{H}(\vec{r}) \psi_m(\vec{r})\,,
\end{aligned} \tag{11}$$

where in the last step Eq. (9) was used, demonstrating that the energy is a $\vec{k}$-dependent quantity.

To further simplify the above expression, matrix elements of the Hamiltonian must be defined. Combined with the overlap integrals introduced earlier, there are three groups of matrix elements. The first group is the atomic energy shift due to the effects of the crystal potential:

$$\beta_m = -\int \mathrm{d}^3\vec{r} \phi_m^*(\vec{r}) \Delta\hat{H} \phi_m(\vec{r})\,. \tag{12}$$

The more general form of the overlap integral is evaluated between different orbitals of neighbouring atoms [7]:

$$\alpha_{m,l}(\vec{R}_p) = \int \mathrm{d}^3\vec{r} \phi_m^*(\vec{r}) \phi_l(\vec{r}-\vec{R}_p)\,. \tag{13}$$

These first two terms are typically small in comparison with the third term, the interatomic matrix element:

$$\gamma_{m,l}(\vec{R}_p) = -\int \mathrm{d}^3\vec{r}\phi_m^*(\vec{r})\Delta\hat{H}\phi_l(\vec{r}-\vec{R}_p)\,. \tag{14}$$

This dominant contribution is commonly referred to as the bond energy [6]. The final form of the energy, expressed with the quantities introduced above, is:

$$\varepsilon_m(\vec{k}) = E_m - \beta_m + \sum_{\vec{R}_n\neq 0}\sum_l \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}_n}\gamma_{m,l}(\vec{R}_n)\,, \tag{15}$$

where E_m is the energy of a single atomic level and $a_m^*(\vec{0})a_m(\vec{0}) \approx \frac{1}{N}$, since the overlap integrals are assumed to be zero.

2.1.1 Wannier states

However, before converting to a discrete lattice formalism, one must address the problem of non-orthogonal free atomic orbitals. If these are not orthogonal, one cannot assume that the overlap integrals are exactly zero; therefore, a more suitable basis should be chosen. A very natural choice in solid-state physics is the set of Wannier functions, which form a complete orthogonal basis for localised orbitals in a crystalline system. These Wannier states are created by performing a Fourier transform on the Bloch states $\psi_{n,\vec{k}}(\vec{r})$ belonging to band n,and utilising an appropriate gauge to achieve orthogonality. For a Bloch state $\psi_{n,\vec{k}}(\vec{r})$ in band n with wave vector $\vec{k}$, the corresponding Wannier function centred at lattice site $\vec{R}_m$ is given by:

$$W_n(\vec{r}-\vec{R}_m) = \frac{1}{\sqrt{N}}\sum_{\vec{k}} \mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{R}_m}\psi_{n,\vec{k}}(\vec{r})\,, \tag{16}$$

where the summation runs over the first Brillouin zone. The resulting states are orthonormal for different sites and different bands:

$$\int \mathrm{d}^3\vec{r}\, W_m^*(\vec{r}-\vec{R}_i)W_n(\vec{r}-\vec{R}_j) = \delta_{mn}\delta_{ij}\,. \tag{17}$$

This crucial property allows us to define the creation and annihilation operators acting on the Wannier states while ensuring they obey the standard fermionic anticommutation relations.

2.1.2 Second quantisation and lattice representation

In the remainder of this study, we will focus on two-dimensional square lattices featuring nearest-neighbour spin-orbit interactions. A highly applicable formalism for this task is the second quantisation of the tight-binding model [5].

By projecting the continuous Hamiltonian onto the newly defined orthonormal Wannier basis, we can extract the specific parameters needed for our discrete grid. In a discrete lattice representation, the on-site energy is associated with the difference between the single atomic energy level and the atomic energy shift at lattice site m:

$$E_m - \beta_m \longleftrightarrow \epsilon_m \,, \tag{18}$$

and the interatomic matrix element corresponds to the hopping amplitude between sites m and l:

$$\gamma_{m,l}(\vec{R}_p) \longleftrightarrow t_{ml} \,. \tag{19}$$

Here, the explicit $\vec{R}_p$ dependence of the hopping amplitude is omitted for brevity, as any spatial dependence of the hopping amplitudes will be inherently denoted via the site indices. Thus, the second-quantised Hamiltonian takes the form:

$$\hat{H} = \sum_{m,\sigma} \epsilon_m \hat{c}^{\dagger}_{m,\sigma} \hat{c}_{m,\sigma} + \sum_{\langle m,l \rangle,\sigma} \left(t_{ml} \hat{c}^{\dagger}_{m,\sigma} \hat{c}_{l,\sigma} + \text{h.c.} \right) \,, \tag{20}$$

where $\hat{c}^{\dagger}_{m,\sigma}$ and $\hat{c}_{m,\sigma}$ are spin-polarised creation and annihilation operators, and h.c. stands for Hermitian conjugate. The sum over $\langle m, l \rangle$ runs strictly over the nearest-neighbour pairs of the lattice, while σ dictates the spin polarisations.

2.1.3 Single-particle matrix formalism

Because the specific Hamiltonian of interest does not include explicit electron-electron interactions, the many-body problem can be successfully mapped to a single-particle matrix eigenvalue problem. Therefore, we can transition from the second-quantised operators back to a single-particle bra-ket notation to construct the block matrices we will numerically diagonalise.

Moving forward, we will operate on a square grid with N_x and N_y sites in their respective directions. Each site will possess an internal (spin) degree of freedom and an external (orbital or atomic type) degree of freedom. To explicitly reflect the underlying grid geometry alongside these degrees of freedom, a new localised basis set of states can be defined:

$$|\Psi^{j,\alpha}_{m_x,m_y}\rangle = |m_x, m_y\rangle \otimes |j\rangle \otimes |\alpha\rangle \,, \tag{21}$$

where $|m_x, m_y\rangle$ describes the site position in the lattice, and $|j\rangle$ and $|\alpha\rangle$ represent the orbital and spin degrees of freedom, respectively. Finally, we reach the explicit matrix form of the Hamiltonian:

$$\begin{aligned}\hat{H} = &\sum_{m_x,m_y}\sum_{j,\alpha}\sum_{l,\beta} U_{j\alpha,l\beta}\ |\Psi^{j,\alpha}_{m_x,m_y}\rangle\langle\Psi^{l,\beta}_{m_x,m_y}| \\ &+\frac{1}{2}\sum_{m_x,m_y}\sum_{\delta\in\mathcal{N}}\sum_{j,\alpha}\sum_{l,\beta}\left(T_\delta^{j\alpha,l\beta}\ |\Psi^{j,\alpha}_{m_x+\delta_x,m_y+\delta_y}\rangle\langle\Psi^{l,\beta}_{m_x,m_y}| + \text{h.c.}\right),\end{aligned} \tag{22}$$

where the factor of $1/2$ corrects for double-counting, δ represents a neighbour within the collection of nearest neighbours $\mathcal{N}$, $U_{j\alpha,l\beta}$ are the on-site matrix elements, and $T_\delta^{j\alpha,l\beta}$ are the direction-dependent hopping amplitudes. After performing the summation over the internal spin and orbital degrees of freedom, one obtains the hopping and on-site operators acting strictly on those subspaces:

$$\sum_{j,\alpha}\sum_{l,\beta} U_{j\alpha,l\beta}\ |\Psi^{j,\alpha}_{m_x,m_y}\rangle\langle\Psi^{l,\beta}_{m_x,m_y}| = \hat{U}\otimes|m_x,m_y\rangle\langle m_x,m_y|\ , \tag{23}$$

$$\sum_{j,\alpha}\sum_{l,\beta} T_\delta^{j\alpha,l\beta}\ |\Psi^{j,\alpha}_{m_x+\delta_x,m_y+\delta_y}\rangle\langle\Psi^{l,\beta}_{m_x,m_y}| = \hat{T}_\delta\otimes|m_x+\delta_x,m_y+\delta_y\rangle\langle m_x,m_y|\ . \tag{24}$$

This allows the Hamiltonian to be compactly written as:

$$\begin{aligned}\hat{H} = &\sum_{m_x,m_y}\hat{U}\otimes|m_x,m_y\rangle\langle m_x,m_y| \\ &+\frac{1}{2}\sum_{m_x,m_y}\sum_{\delta\in\mathcal{N}}\left(\hat{T}_\delta\otimes|m_x+\delta_x,m_y+\delta_y\rangle\langle m_x,m_y| + \text{h.c.}\right).\end{aligned} \tag{25}$$

In the case of strict translational invariance across the lattice, we can express the system in momentum space. For a periodic lattice, the real-space basis states can be discrete Fourier transformed into momentum-space states:

$$|\Psi^{j,\alpha}_{\vec{k}}\rangle = \frac{1}{\sqrt{N_xN_y}}\sum_{m_x,m_y} \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}_{m_x,m_y}}\ |\Psi^{j,\alpha}_{m_x,m_y}\rangle\ . \tag{26}$$

Applying this transformation, the obtained Bloch Hamiltonian is:

$$\hat{H}(\vec{k}) = \hat{U} + \sum_{\delta\in\mathcal{N}}\left(\hat{T}_\delta\mathrm{e}^{-\mathrm{i}\vec{k}\cdot\vec{\delta}} + \hat{T}_\delta^\dagger\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{\delta}}\right)\ . \tag{27}$$

2.2 Heisenberg model

In order to describe magnetic properties quantum-mechanically, one may use the quantum Heisenberg model [2, 3, 8]. This statistical model describes magnetism through interactions of localised spin sites. Looking at an n-dimensional lattice, at each site i, a molecule (or any object with a localised magnetic property) with spin operator $\vec{S}_i$ is placed. Considering only the simplest two-spin interactions, the following Hamiltonian can be formulated:

$$\hat{H} = \frac{1}{2} \sum_{i,j} J_{ij} \vec{S}_i \cdot \vec{S}_j \,, \tag{28}$$

where J_{ij} are the exchange constants and the summation is performed over all interacting instances.

The exchange constant originates from the microscopic interactions between the localised electronic wave-functions. In the simplest case, this can be expressed as an exchange integral in a Coulomb potential, but it may also contain other terms, such as superexchange interactions in highly correlated systems [2]. The sign of J_{ij} describes the overall behaviour of the system. Ferromagnets have negative exchange constants, while antiferromagnets have positive ones.

The spin operators follow the usual commutation relation:

$$\left[S_k^\alpha, S_j^\beta \right] = \mathrm{i}\hbar \varepsilon_{\alpha\beta\gamma} \delta_{kj} S_k^\gamma \,, \tag{29}$$

and have the eigenvalue $\vec{S}^2 = \hbar^2 S(S+1)$, where S is an integer or half-integer.

At high temperatures, due to thermal fluctuations, the expectation values of the spins are zero:

$$\langle \vec{S}_i \rangle = \mathrm{Tr} \left(\rho \vec{S}_i \right) = 0 \,, \tag{30}$$

where ρ is the density matrix. However, under a critical temperature T_c, the spins point in a distinguished direction, therefore $\langle \vec{S}_i \rangle \neq 0$.

2.2.1 Ferromagnetic case

The ground state of the ferromagnet is the fully polarised case, in which all spins point in the same direction. In order to prove this, take a lattice with spins of the same size S and eigenvalue $S^z = \hbar m$. The basis is then $|S_i, m_i\rangle = |m_i\rangle$, since all S_i are equal. One can then write the Hamiltonian in the form of:

$$\hat{H} = \frac{1}{2} \sum_{i,j} J_{ij} \left(\frac{S_i^+ S_j^- + S_i^- S_j^+}{2} + S_i^z S_j^z \right) \,, \tag{31}$$

where $S_i^+ = S_i^x + \mathrm{i}S_i^y$ and $S_i^- = S_i^x - \mathrm{i}S_i^y$. Acting with the Hamiltonian on a product state $|\Psi\rangle$ composed of spin eigenstates $|m_i\rangle$ gives:

$$\begin{aligned}\langle\Psi|\,\hat{H}\,|\Psi\rangle &= \frac{1}{2}\sum_{i,j} J_{ij}\left(\langle\Psi|\,\tfrac{S_i^+S_j^-}{2}\,|\Psi\rangle + \langle\Psi|\,\tfrac{S_i^-S_j^+}{2}\,|\Psi\rangle + \langle\Psi|\,S_i^zS_j^z\,|\Psi\rangle\right)\\ &= \frac{1}{2}\sum_{i,j} J_{ij}\,\hbar^2 m_i m_j\,,\end{aligned} \tag{32}$$

since the first two terms vanish when acting on states that are products of S_i^z eigenstates.

In the case of a fully polarised system ($|\Psi_0\rangle$) in the z-direction, all m_i take the value of S, thus:

$$\begin{aligned}\hat{H}\,|\Psi_0\rangle &= \frac{1}{2}\sum_{i,j} J_{ij}\left(\frac{S_i^+S_j^-}{2}\,|\Psi_0\rangle + \frac{S_i^-S_j^+}{2}\,|\Psi_0\rangle + S_i^zS_j^z\,|\Psi_0\rangle\right)\\ &= \frac{1}{2}\sum_{i,j} J_{ij}S_i^zS_j^z\,|\Psi_0\rangle = \frac{\hbar^2S^2}{2}\sum_{i,j} J_{ij}\,|\Psi_0\rangle\ .\end{aligned} \tag{33}$$

Since all J_{ij} are negative, $\hbar^2S^2J_{ij} \leq \hbar^2 m_i m_j J_{ij}$ is true, and $|\Psi_0\rangle$ is an eigenstate of $\hat{H}$. Therefore, the fully polarised case is truly the ground state of the ferromagnetic system. With the definition of total spin $\vec{S} = \sum_i \vec{S}_i$, the ground state has a degeneracy of $2S_{tot}+1$. The ground state also breaks the rotational symmetry of the Hamiltonian (a single unique direction is chosen for the polarisation), this is called spontaneous symmetry breaking.

The simplest excited state is one with a single spin flipped relative to the ground state:

$$|\Psi_n\rangle = S_n^-\,|\Psi_0\rangle\ . \tag{34}$$

However, this is not an eigenstate of $\hat{H}$, but a superposition of such states is:

$$|\phi\rangle = \frac{1}{\sqrt{N}}\sum_n \mathrm{e}^{\mathrm{i}kn}S_n^-\,|\Psi_0\rangle\ . \tag{35}$$

These states are called spin-waves or magnons. Subsequent excitations are multi-magnon states:

$$|\Psi_{multi}\rangle = \left(\prod_i S_i^-\right)|\Psi_0\rangle\ . \tag{36}$$

2.2.2 Antiferromagnetic case

Exact solutions for the antiferromagnetic ground state exist only in a few cases [2, 3, 8]. One case, which has some rigorous results, is the bipartite lattice. In such a lattice, two distinct sublattices exist, denoted A and B, that have bonds connecting sites from different sublattices only. A two-dimensional example is a chequerboard lattice with alternating sites from A and B. One might think that the ground state for this setting is two fully polarised sublattices with opposing directions, called the Néel state. However, this is not the true ground state. The true ground state is a singlet state with the total spin $S_{tot} = 0$, according to Marshall's theorem [2]. In such states, quantum fluctuations (magnon corrections) mix spin configurations and lower the energy relative to the classical Néel arrangement. Due to fluctuations, long-range order exists only in three dimensions, and in two dimensions at zero temperature [3, 8]. Finite-temperature two-dimensional and one-dimensional systems have no long-range order.

2.2.3 Relativistic extensions and Classical Mapping

The isotropic Heisenberg model successfully describes the collinear alignment of spins driven by the Coulomb interaction and the Pauli exclusion principle. However, it assumes full rotational invariance in spin space, treating the spins as completely decoupled from the underlying crystal lattice. In real materials, strong spin-orbit coupling (SOC) connects the spin degrees of freedom to the spatial orbital motion of the electrons.

The combination of strong SOC and broken inversion symmetry makes the expansion of the spin Hamiltonian to include relativistic, anisotropic magnetic interactions necessary. The most general form of the bilinear interaction [4] between two localised magnetic moments can be written using a general 3×3 exchange tensor $\mathcal{J}_{ij}$:

$$\hat{H}_{\text{bilinear}} = \sum_{i \neq j} \vec{S}_i \cdot \mathcal{J}_{ij} \cdot \vec{S}_j \,. \tag{37}$$

This generic interaction tensor can be decomposed into three components: a scalar trace, an antisymmetric matrix, and a traceless symmetric matrix:

$$\mathcal{J}_{ij} = J_{ij}\mathbb{I} + \mathcal{J}_{ij}^{\text{A}} + \mathcal{J}_{ij}^{\text{S}} \,, \tag{38}$$

where $J_{ij} = \frac{1}{3}\text{Tr}(\mathcal{J}_{ij})$ is the isotropic Heisenberg exchange constant, which favours collinearity. The antisymmetric part, $\mathcal{J}_{ij}^{\text{A}} = \frac{1}{2}(\mathcal{J}_{ij} - \mathcal{J}_{ij}^{T})$, can be mapped to an axial vector $\vec{D}_{ij}$, yielding the Dzyaloshinskii-Moriya interaction

(DMI) in the form of $\vec{D}_{ij} \cdot (\vec{S}_i \times \vec{S}_j)$. Unlike the isotropic exchange, due to the cross-product in the DMI, this exchange energetically favours orthogonal spin alignments. The traceless symmetric tensor $\mathcal{J}_{ij}^{\mathrm{S}} = \frac{1}{2}(\mathcal{J}_{ij} + \mathcal{J}_{ij}^{T}) - J_{ij}\mathbb{I}$ is usually denoted as $\bar{\bar{\Gamma}}_{ij}$ and represents the symmetric anisotropic exchange, acting as a highly directional two-site coupling. Furthermore, a single-ion magnetic anisotropy term, governed by a local symmetric tensor $\mathcal{K}_i$, is often added to account for the crystal electric field and second-order SOC effects.

In modern computational approaches, it is common to treat the magnetic moments semi-classically to evaluate the energy of complex magnetic configurations. In the classical limit, the spin operators are replaced by dimensionless classical unit vectors $\vec{e}_i$ using the mapping $\vec{S}_i \to \hbar S_i \vec{e}_i$, where S_i is the spin quantum number of site i.

To formulate the classical energy, the interaction parameters are renormalised to have energy units by absorbing the spin magnitudes. For example, the full 3×3 interaction tensor is scaled as $\mathcal{J}_{ij}^{\mathrm{cl}} = \hbar^2 S_i S_j \mathcal{J}_{ij}$ and the single-ion magnetic anisotropy tensor as $K_i^{\mathrm{cl}} = (\hbar S_i)^2 \mathcal{K}_i$.

When decomposing the bilinear interaction into its isotropic (J_{ij}^{cl}), antisymmetric Dzyaloshinskii-Moriya ($\vec{D}_{ij}^{\mathrm{cl}}$), and symmetric anisotropic ($\bar{\bar{\Gamma}}_{ij}^{\mathrm{cl}}$) components, these individual terms carry the absorbed $\hbar^2 S_i S_j$ scalar factor.

Furthermore, the single-ion magnetic anisotropy term, governed by the local symmetric tensor K_i^{cl}, is added to account for the crystal electric field and second-order SOC effects. This on-site tensor breaks the continuous rotational symmetry by pinning the macroscopic magnetisation to specific crystallographic directions, resulting in magnetic easy axes or easy planes. Thus, the full classical extended Heisenberg Hamiltonian takes the form:

$$H = \frac{1}{2}\sum_{i\neq j} J_{ij}^{\mathrm{cl}} \vec{e}_i \cdot \vec{e}_j + \frac{1}{2}\sum_{i\neq j} \vec{D}_{ij}^{\mathrm{cl}} \cdot (\vec{e}_i \times \vec{e}_j) + \frac{1}{2}\sum_{i\neq j} \vec{e}_i \cdot \bar{\bar{\Gamma}}_{ij}^{\mathrm{cl}} \cdot \vec{e}_j + \sum_i \vec{e}_i \cdot K_i^{\mathrm{cl}} \cdot \vec{e}_i \,. \quad (39)$$

The factor of $1/2$ corrects for the double-counting in the summation.

2.3 Bernevig-Hughes-Zhang model

The Bernevig-Hughes-Zhang (BHZ) model was first introduced to provide an accurate low-energy description of a two-dimensional topological insulator. The most prominent characteristics of such systems are strong spin-orbit coupling (inherent to heavy elements like Hg and Te), potential band inversion, and robust propagating edge states crossing the band gap. The first experimental realisation of the model was an HgTe layer in a CdTe quantum well [1].

In a standard semiconductor, the s-type conduction band sits at a higher energy level than the p-type valence band. However, in HgTe, strong relativistic effects pull the p-like bands above the s-like bands. By surrounding a layer of HgTe with CdTe (a normal insulator with a band gap at a different energy level, thus creating a quantum well), the electronic properties can be tuned by the thickness d of the HgTe layer. The lowest conduction sub-band ($E1$) and the highest valence sub-band ($H1$) cross and invert at a critical thickness ($d > d_c \approx 6.3$ nm), transforming the system into a topological phase (propagating edge states appear).

Looking at the bulk electronic structure of HgTe, it is characterised by eight energy bands near the Fermi level, originating from the s-like (Γ_6) and p-like (Γ_8 and Γ_7) atomic orbitals. A realistic model of this system is formally derived from a $\vec{k}\cdot\vec{p}$ calculation starting with a six-band (or eight-band) Kane model [9, 10]. However, due to quantum confinement (the energy level of the light-hole Γ_8 bands are pushed away due to the small effective mass) and sufficient energy separation in the band structure (Γ_7 spin-orbit split-off bands have a 1 eV gap), four of these bands can be neglected for the tight-binding formulation [10, 11].

Consequently, in the low-energy limit near the Γ point ($\vec{k} = 0$), the system is projected onto a four-band effective model dominated by the remaining four states: the spin-up and spin-down components of the $E1$ and $H1$ bands. Thus, our basis is defined as:

$$\Psi_{\vec{k}} = \left(|E1,+\rangle, |H1,+\rangle, |E1,-\rangle, |H1,-\rangle\right)^T , \tag{40}$$

where $\pm$ denote the spin projections. Due to the presence of time-reversal symmetry ($\mathcal{T}$) and spatial inversion symmetry in the unperturbed system, the effective Hamiltonian takes a block-diagonal form:

$$\mathcal{H}_{\mathrm{BHZ}}(\vec{k}) = \begin{pmatrix} h(\vec{k}) & 0 \\ 0 & h^*(-\vec{k}) \end{pmatrix} . \tag{41}$$

The upper 2×2 block $h(\vec{k})$ describes the spin-up sector, while the lower block $h^*(-\vec{k})$ describes the spin-down sector. Due to the presence of time-reversal symmetry ($\mathcal{T}$) and spatial inversion symmetry, these two blocks are mathematically related to each other. The Hamiltonian for a single spin block can be expanded using the Pauli matrices $\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$ acting on the orbital $(E1, H1)$ space:

$$h(\vec{k}) = \epsilon(\vec{k})\hat{\sigma}_0 + \vec{d}(\vec{k}) \cdot \hat{\boldsymbol{\sigma}} , \tag{42}$$

where $\hat{\sigma}_0$ is the 2×2 identity matrix in the orbital subspace. The specific momentum-dependent components, preserving the relevant crystal symmetries up to the second order in $\vec{k}$, are given by:

$$\epsilon(\vec{k}) = C - D(k_x^2 + k_y^2)\,, \tag{43}$$

$$d_x(\vec{k}) = Ak_x\,, \tag{44}$$

$$d_y(\vec{k}) = -Ak_y\,, \tag{45}$$

$$d_z(\vec{k}) = M - B(k_x^2 + k_y^2)\,. \tag{46}$$

The structure of this single spin block $h(\vec{k})$ is analogous to the relativistic Dirac equation in 2+1 dimensions. In this context, the linear momentum terms (Ak_x and $-Ak_y$) describe massless Dirac fermions, where the parameter A represents the Fermi velocity ($A \approx \hbar v_F$), acting analogously to the speed of light. The parameter M represents the effective Dirac mass, which introduces a gap in the energy spectrum. Furthermore, the quadratic $B(k_x^2 + k_y^2)$ term ensures the correct behaviour at large momenta in the Brillouin zone.

Therefore, the topological phase transition, driven by the tuning of the thickness of the HgTe layer, is equivalent to a Dirac mass inversion. The sign of the fraction M/B determines the topological state of the system. When $M/B < 0$ (the thin quantum well regime), the bands are in their normal order, yielding a trivial insulator with a conventional band gap. On the other hand, when $M/B > 0$ (the thick quantum well regime), the $E1$ and $H1$ bands are inverted. In this case, the system is a $\mathbb{Z}_2$ non-trivial topological insulator characterised by an insulating bulk but hosting gapless, counter-propagating, and spin-momentum locked edge states.

2.3.1 Time-reversal and Chiral Symmetries

The fundamental reasons for the topological protection of the BHZ model and the existence of the Quantum Spin Hall (QSH) phase are the underlying symmetries of the Hamiltonian. The most critical of these is the time-reversal symmetry ($\mathcal{T}$).

For a spin-$\frac{1}{2}$ fermionic system, the time-reversal operator ($\mathcal{T}$) is anti-unitary and satisfies $\mathcal{T}^2 = -1$. In the chosen basis of the BHZ model, it can be represented as $\mathcal{T} = -\mathrm{i}\hat{s}_y\mathcal{K}$, where $\hat{s}_y$ acts on the spin subspace and $\mathcal{K}$ denotes complex conjugation. The BHZ Hamiltonian is invariant under time reversal, meaning:

$$\mathcal{T}\mathcal{H}_{\mathrm{BHZ}}(\vec{k})\mathcal{T}^{-1} = \mathcal{H}_{\mathrm{BHZ}}(-\vec{k})\,. \tag{47}$$

Because spin-mixing interactions are not present, the Hamiltonian assumes a 4×4 block diagonal form, in which the lower and upper 2×2 blocks are

time-reversed partners. According to Kramers' theorem, every energy eigenstate in a $\mathcal{T}$-invariant system must have a time-reversed partner at $-\vec{k}$. At the time-reversal invariant momenta (TRIMs), this forces the states to be degenerate in energy. At the physical boundaries of the sample, this symmetry ensures the existence of a Kramers pair of counter-propagating edge states with opposite spins. As long as time-reversal symmetry is preserved (i.e., no magnetic impurities or external magnetic fields are present), these helical edge states are protected against elastic single-particle backscattering. A $\mathcal{T}$-symmetric, non-magnetic defect cannot scatter an electron into its counter-propagating Kramers partner. While inelastic scattering (such as by phonons) or many-body interactions can still introduce dissipation at finite temperatures, this topological protection against elastic backscattering makes the edge states robust.

Another important symmetry to consider is Chiral Symmetry. A Hamiltonian possesses chiral symmetry if there exists a unitary operator $\mathcal{C}$ that anti-commutes with the Hamiltonian, meaning $\{\mathcal{H}, \mathcal{C}\} = 0$.

If we consider the massless limit of the BHZ model where $M = 0$ alongside $C = 0$ and $D = 0$, the remaining terms $(Ak_x\hat{\sigma}_x - Ak_y\hat{\sigma}_y)$ anti-commute with the operator $\mathcal{C} = \hat{\sigma}_z$. In this idealised limit, the system possesses exact chiral symmetry.

However, introducing the Dirac mass $M\hat{\sigma}_z$ breaks this chiral symmetry. Because the effective $\vec{d}(\vec{k})$ vector now spans all three Pauli matrices, there is no constant orthogonal operator left that can anti-commute with the entire Hamiltonian.

Despite this breaking of chiral symmetry, as long as C and D are zero, the simplified massive Dirac energy spectrum remains symmetric around zero: $E(\vec{k}) = \pm\sqrt{A^2k^2 + M^2}$. This is simply a consequence of the Hamiltonian being traceless, rather than a result of chiral symmetry.

In the realistic BHZ model for HgTe/CdTe, the terms C and $D(k_x^2 + k_y^2)$ are non-zero. The parameter C acts as a simple global energy shift, while the D term introduces true electron-hole asymmetry (yielding different effective masses for the conduction and valence bands). Since these terms are proportional to the identity matrix $\hat{\sigma}_0$ in the orbital subspace, they give the Hamiltonian a non-zero trace, bending the energy bands and destroying the $\pm E$ symmetry.

It is worth noting, that while the breaking of chiral symmetry bends the energy bands, it does not destroy the topological phase. The $\mathbb{Z}_2$ topological invariant relies on the preservation of time-reversal symmetry. However, introducing a magnetic field (such as coupling the BHZ layer to a magnetic material) will break $\mathcal{T}$, lifting the Kramers degeneracy and driving the system into new phases, such as the Quantum Anomalous Hall (QAH) insulator.

2.3.2 Discretization: The Real-Space Tight-Binding Hamiltonian

While the continuous $\vec{k} \cdot \vec{p}$ Hamiltonian provides physical insight into the topological phase transition, large-scale numerical simulations (such as evaluating the real-space magnetic interactions in a heterostructure) require a discrete lattice model. The continuous BHZ Hamiltonian can be regularised onto a two-dimensional square lattice by replacing the continuous momentum operators with their periodic equivalents. Assuming a lattice constant of $a = 1$, the linear and quadratic terms in the Brillouin zone are substituted as:

$$k_i \longrightarrow \sin(k_i)\,, \tag{48}$$
$$k_i^2 \longrightarrow 2(1 - \cos(k_i))\,, \tag{49}$$

where $i \in \{x, y\}$. These substitutions preserve the parity of the original continuous functions. The linear, odd k_i terms are replaced by the odd $\sin(k_i)$ functions, while the quadratic, even k_i^2 terms are replaced by even cosine functions. This mapping ensures that the fundamental spatial inversion and time-reversal symmetries of the system remain intact on the discrete lattice, while preventing unphysical fermion doubling at the Brillouin zone boundaries.

After applying the previous substitutions, the spin-up block becomes $h_\uparrow(\mathbf{k}) = \epsilon(\mathbf{k})\hat{\sigma}_0 + \mathbf{d}(\mathbf{k}) \cdot \hat{\boldsymbol{\sigma}}$, and the regularised momentum-space Bloch Hamiltonian becomes:

$$\begin{aligned} h_\uparrow(\mathbf{k}) = \; & [C - 2D(2 - \cos k_x - \cos k_y)]\,\hat{\sigma}_0 \\ & + A \sin k_x \hat{\sigma}_x - A \sin k_y \hat{\sigma}_y \\ & + [M - 2B(2 - \cos k_x - \cos k_y)]\,\hat{\sigma}_z\,. \end{aligned} \tag{50}$$

To translate this into a real-space lattice model, we group the terms by their spatial dependencies:

$$\begin{aligned} h_\uparrow(\mathbf{k}) = & (C - 4D)\hat{\sigma}_0 + (M - 4B)\hat{\sigma}_z \\ & + 2D \cos k_x \hat{\sigma}_0 + A \sin k_x \hat{\sigma}_x + 2B \cos k_x \hat{\sigma}_z \\ & + 2D \cos k_y \hat{\sigma}_0 - A \sin k_y \hat{\sigma}_y + 2B \cos k_y \hat{\sigma}_z\,. \end{aligned} \tag{51}$$

The transformation to real space is achieved by expressing the trigonometric functions via Euler's formula, and by isolating the coefficient of the forward-hopping phases, we extract the nearest-neighbour hopping and on-

site potential matrices:

$$\hat{t}_{x,\uparrow} = D\hat{\sigma}_0 + B\hat{\sigma}_z + \mathrm{i}\frac{A}{2}\hat{\sigma}_x\,, \tag{52}$$

$$\hat{t}_{y,\uparrow} = D\hat{\sigma}_0 + B\hat{\sigma}_z - \mathrm{i}\frac{A}{2}\hat{\sigma}_y\,, \tag{53}$$

$$\hat{u}_{\uparrow} = (C - 4D)\hat{\sigma}_0 + (M - 4B)\hat{\sigma}_z\,. \tag{54}$$

Due to the preserved time-reversal symmetry, the spin-down block $h_{\downarrow}(\mathbf{k})$ is the Kramers partner of the spin-up block, satisfying $h_{\downarrow}(\mathbf{k}) = h^*_{\uparrow}(-\mathbf{k})$. Thus, the spin down components are:

$$\hat{t}_{x,\downarrow} = D\hat{\sigma}_0 + B\hat{\sigma}_z - \mathrm{i}\frac{A}{2}\hat{\sigma}_x\,, \tag{55}$$

$$\hat{t}_{y,\downarrow} = D\hat{\sigma}_0 + B\hat{\sigma}_z - \mathrm{i}\frac{A}{2}\hat{\sigma}_y\,, \tag{56}$$

$$\hat{u}_{\downarrow} = (C - 4D)\hat{\sigma}_0 + (M - 4B)\hat{\sigma}_z\,. \tag{57}$$

Consequently, the full 4×4 real-space matrices for the internal orbital subspace are block-diagonal:

$$\hat{U} = \begin{pmatrix} \hat{u}_{\uparrow} & 0 \\ 0 & \hat{u}_{\downarrow} \end{pmatrix}, \quad \hat{T}_x = \begin{pmatrix} \hat{t}_{x,\uparrow} & 0 \\ 0 & \hat{t}_{x,\downarrow} \end{pmatrix}, \quad \hat{T}_y = \begin{pmatrix} \hat{t}_{y,\uparrow} & 0 \\ 0 & \hat{t}_{y,\downarrow} \end{pmatrix}. \tag{58}$$

Following the general formalism for a finite 2D lattice of size $N_x \times N_y$, we can now construct the most general form of the real-space tight-binding Hamiltonian:

$$\begin{aligned} \hat{H}_{\mathrm{TB}} = &\sum_{m_x=1}^{N_x}\sum_{m_y=1}^{N_y}\left(\hat{U}\left|\Psi_{m_x,m_y}\right\rangle\left\langle\Psi_{m_x,m_y}\right|\right) \\ &+ \sum_{m_x=1}^{N_x-1}\sum_{m_y=1}^{N_y}\left(\hat{T}_x\left|\Psi_{m_x+1,m_y}\right\rangle\left\langle\Psi_{m_x,m_y}\right| + \mathrm{h.c.}\right) \\ &+ \sum_{m_x=1}^{N_x}\sum_{m_y=1}^{N_y-1}\left(\hat{T}_y\left|\Psi_{m_x,m_y+1}\right\rangle\left\langle\Psi_{m_x,m_y}\right| + \mathrm{h.c.}\right)\,, \end{aligned} \tag{59}$$

where h.c. denotes the Hermitian conjugate. The state vector $|\Psi_{m_x,m_y}\rangle$ separates the lattice supercell position, the orbital degree of freedom within the cell, and the internal spin degree of freedom:

$$\left|\Psi_{m_x,m_y}\right\rangle = \left|m_x, m_y\right\rangle \otimes \left|j\right\rangle \otimes \left|\alpha\right\rangle\,. \tag{60}$$

Here, $|m_x, m_y\rangle$ describes the position of the cell in the lattice, $|j\rangle$ describes the tight-binding orbital sites ($j = E1, H1$), and $|\alpha\rangle$ fixes the internal spin basis ($|\uparrow\rangle$ or $|\downarrow\rangle$).

2.3.3 Symmetry Considerations and the Doubled Chern Insulator

An instructive way to view this constructed real-space BHZ model is to frame it as two decoupled copies of a two-dimensional Chern insulator. Following the formalism presented here [12], the BHZ model can be viewed as a time-reversal invariant extension of the Qi-Wu-Zhang (QWZ) model.

By assigning one copy of the QWZ model to the spin-up electrons and a time-reversed copy to the spin-down electrons, the comprehensive 4×4 Bloch Hamiltonian is formulated using the tensor product between the internal spin subspace (spanned by $\hat{s}_{x,y,z}$ and the identity $\hat{s}_0$) and the orbital subspace (spanned by $\hat{\sigma}_{x,y,z}$ and the identity $\hat{\sigma}_0$):

$$\hat{H}_{\mathrm{BHZ}}(\mathbf{k}) = \hat{s}_0 \otimes [(u + \cos k_x + \cos k_y)\hat{\sigma}_z + \sin k_y \hat{\sigma}_y] + \hat{s}_z \otimes \sin k_x \hat{\sigma}_x + \hat{s}_x \otimes \hat{\Lambda}\,. \tag{61}$$

Here, u is a tunable dimensionless mass parameter, and $\hat{\Lambda}$ represents a Hermitian spin-mixing coupling term. Assuming no spin-mixing mechanism in the pure lattice ($\hat{\Lambda} = 0$), the Hamiltonian falls apart into the two uncoupled blocks derived previously.

Mapping Material Parameters to the Toy Model

Although the previously described toy model captures the exact topological phase transitions using a single parameter u, the numerical implementation that describes the experimental results utilises realistic material parameters (A, B, C, D, M). It is instructive to establish the correspondence between these two formalisms.

First, we can neglect two terms by setting $C = 0$ and $D = 0$. These terms only introduce a global energy shift and bend the bands asymmetrically, but they do not open or close the band gap and therefore do not influence the topological invariant. Second, the lattice constant is normalised to $a = 1$, and the hopping amplitudes are scaled to unity by setting $A = 1$ and $B = 1/2$.

The final mapping involves the Dirac mass M, which drives the topological phase transition. In our realistic model discretised with $B = 1/2$, the diagonal mass term takes the form:

$$d_z(\mathbf{k}) = M - 2\left(\frac{1}{2}\right)(2 - \cos k_x - \cos k_y) = (M - 2) + \cos k_x + \cos k_y\,. \tag{62}$$

Comparing this to the diagonal term of the toy model, $d_z(\mathbf{k}) = u + \cos k_x + \cos k_y$, we obtain the exact correspondence for the mass parameter:

$$u = M - 2\,. \tag{63}$$

Finally, we should mention the topological phase boundaries of the insulator with respect to the u mass parameter. The insulator is in its topological phase if $u \in (-2, 0) \cup (0, 2)$ and is in its trivial phase if $u \in (-\infty, -2) \cup (2, \infty)$. At the phase boundaries, the band gap closes. Specifically, for $u = 0$, the gap closes at the $\vec{k} = (\pi, 0)$ and $(0, \pi)$ points, whereas for $u = -2$ and $u = 2$, it closes at the Γ and M points, respectively [12].

3 Methods

3.1 Real-Space Hamiltonian

Following the previously established tight-binding formalism, one can now define the Hamiltonian of the studied system. This system consists of a two-dimensional topological insulator (TI) and a coupled magnetic layer. For the modelling of the magnetic layer, we will introduce an additional localised magnetic orbital subject to a local magnetic exchange field:

$$\begin{aligned}\hat{H} = &\sum_{m_x=1}^{N_x}\sum_{m_y=1}^{N_y}\left(\hat{U}\otimes|m_x,m_y\rangle\langle m_x,m_y|\right)\\ &+\sum_{m_x=1}^{N_x-1}\sum_{m_y=1}^{N_y}\left(\hat{T}_x\otimes|m_x+1,m_y\rangle\langle m_x,m_y|+\text{h.c.}\right)\\ &+\sum_{m_x=1}^{N_x}\sum_{m_y=1}^{N_y-1}\left(\hat{T}_y\otimes|m_x,m_y+1\rangle\langle m_x,m_y|+\text{h.c.}\right),\end{aligned} \tag{64}$$

where $|m_x, m_y\rangle$ represents the spatial lattice site. The operators $\hat{U}$, $\hat{T}_x$, and $\hat{T}_y$ act on the internal state space spanned by two spin ($|\alpha\rangle$) and three orbital degrees of freedom ($|j\rangle$) — two of the orbitals correspond to the topological layer and one to the magnetic layer. Consequently, the on-site operator can be expressed on the orbital basis as:

$$\hat{U} = \begin{bmatrix} u\hat{s}_0 & 0 & c_{\text{inter}}\hat{s}_0 \\ 0 & -u\hat{s}_0 & c_{\text{inter}}\hat{s}_0 \\ c_{\text{inter}}\hat{s}_0 & c_{\text{inter}}\hat{s}_0 & \vec{b}\cdot\vec{\boldsymbol{s}}+\mu\hat{s}_0 \end{bmatrix}, \tag{65}$$

where $\hat{s}_0$ acts on the spin subspace. The tunable parameters of the model are the on-site mass term (u), the coupling strength between the magnetic and TI layers (c_{inter}), the magnetic exchange field vector ($\vec{b}$) applied to the magnetic layer, and lastly, the chemical potential shift for the magnetic block. It should be further emphasised that $\vec{b}$ does not describe an external magnetic field, but rather the intrinsic magnetic properties of the magnetic layer.

The inter-layer coupling is constructed as a spin-conserving, on-site hybridisation, denoted by its proportionality to the spin identity matrix $\hat{s}_0$. This ensures that the hopping of electrons between the heterostructure layers does not flip their spins. As a consequence, all time-reversal symmetry breaking is restricted to the exchange field within the magnetic layer itself. Furthermore, this symmetric coupling hybridises both the conduction and

valence orbitals of the TI with the localised magnetic orbital at the exact same spatial lattice site, serving as a phenomenological description of the magnetic proximity effect. Moreover, the hopping terms are defined in a similar way:

$$\hat{T}_x = \frac{1}{2}\begin{bmatrix} \hat{s}_0 & -\mathrm{i}\hat{s}_z & 0 \\ -\mathrm{i}\hat{s}_z & -\hat{s}_0 & 0 \\ 0 & 0 & t_x\hat{s}_0 \end{bmatrix}, \tag{66}$$

$$\hat{T}_y = \frac{1}{2}\begin{bmatrix} \hat{s}_0 & -\hat{s}_0 & 0 \\ \hat{s}_0 & -\hat{s}_0 & 0 \\ 0 & 0 & t_y\hat{s}_0 \end{bmatrix}, \tag{67}$$

where $t_{x/y}$ are the hopping amplitudes along the given direction within the magnetic layer. With these defined operators, the magnetic orbitals affect only those TI orbitals which share the same lattice position. This description is sufficient for the previously discussed toy model. However, for more general capabilities, the actual numerical implementation contains some additional parameters. These parameters are namely the inter-orbital coupling strength within the TI block, the TI normal hopping (effective mass), and the TI spin-orbit hopping (Dirac velocity). The numerical realisation of this Hamiltonian was made by using the `sisl` Python package [13].

The introduction of a magnetic orbital coupled to an exchange field breaks the time-reversal symmetry of the system, thereby rendering the $\mathbb{Z}_2$ topological invariant ill-defined. Instead, the Chern number will be the relevant property of the phase transition. The `Z2Pack` package was used to calculate the Chern number [14, 15].

3.1.1 System parameters

The Hamiltonian of the system in Eq. (64) has many tunable parameters, thus the large parameter space renders the mapping of it rather difficult. To make the latter investigations possible, it must be reduced by setting some parameters constant. Throughout the following calculations, the tunable parameters are the Dirac mass term u, the inter-layer coupling c_{inter}, the exchange field splitting parameter along the z-axis b_z, and the chemical potential shift of the magnetic layer (μ). All the assigned values of the parameters are found in Table 1. From this point forward, the hopping amplitude t_x and t_y are fixed, and their absolute value will act as the unit of measurement, and no energy units will be later indicated. These specific parameters were selected because they form the minimal set required to map the possible phases of the heterostructure. The Dirac mass drives the phase

transition of the isolated TI layer. The exchange field controls the magnitude of the time-reversal symmetry breaking, while the chemical potential dictates the energy alignment between the localised magnetic states and the topological band gap. Finally, the inter-layer coupling (c_{inter}) serves as the interaction knob that hybridises these layers, allowing us to observe how the magnetic proximity effect alters the band structure and drives the system into possible new phases.

Category	**Parameter**	**Symbol**	**Values**
Variable	Inter-layer interaction	c_{inter}	$\{0.1,\ 0.5,\ 1.0\}$
	Magnetic exchange field	$\mathbf{b} = (0, 0, b_z)$	$b_z \in \{0.1,\ 1.0\}$
	Chemical potential	μ	$\{0.0,\ 1.0,\ 1.3,\ 2.0,\ 4.0\}$
	Dirac mass term	u	$\{-2.1,\ -1.9,\ -1.2\}$
Fixed	TI Hopping amplitude	t_{ti}	0.5
	TI Spin-orbit velocity	v_{ti}	0.5
	Intra-layer interaction	c_{intra}	0.0
	Hopping amplitude (x)	t_x	-1.0
	Hopping amplitude (y)	t_y	-1.0

Table 1: Summary of the topological insulator model parameter space.

The parameters are chosen in such a way that the magnetic bands go through several stages of doping in a small and a relatively large magnetic field, and the insulator layer transforms from its topological phase to its trivial case. Having said that, it must be noted that these phases can no longer be categorised as such, since by coupling the magnetic layer to the insulator, the time-reversal symmetry of the insulator layer is broken. However, the band structures still show interesting phenomena in certain combinations of parameters (see Figure 1 and Figure 2). Additionally, the Fermi level is set throughout all cases to make the lower one-third of the bands occupied.

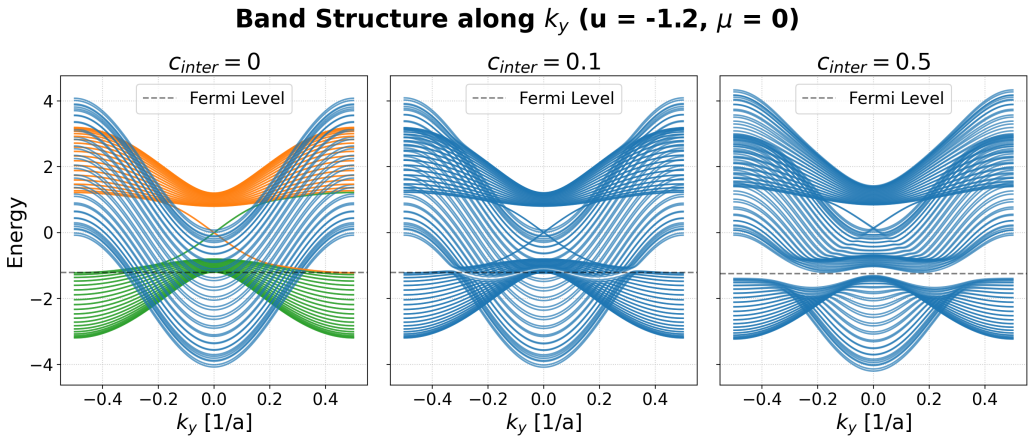


Figure 1: Band structure of the system along the k_y axis. The different colours on the $c_{\text{inter}} = 0$ figure represent different band types. The magnetic bands are coloured blue, while the green and orange bands correspond to the time-reversal symmetric insulator bands. The insulator is in its topological phase. The magnetic field strength is $b_z = 0.1$.

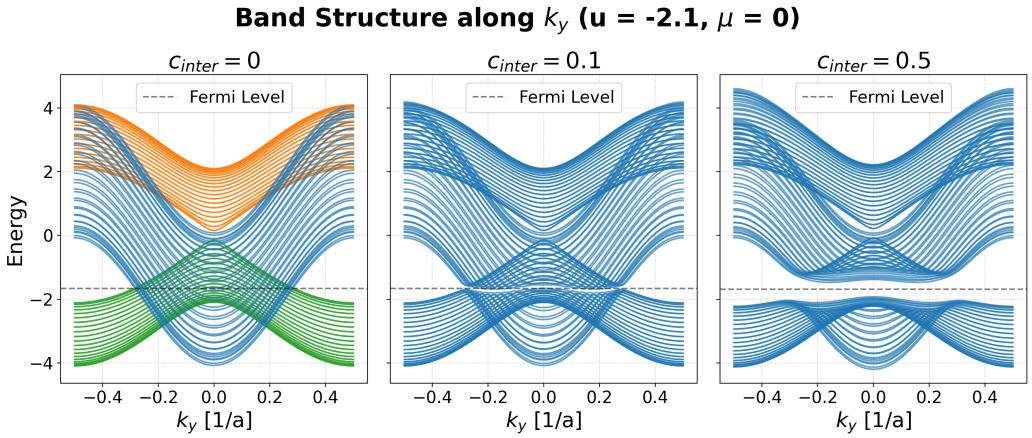


Figure 2: Band structure of the system along the k_y axis. The different colours on the $c_{\text{inter}} = 0$ figure represent different band types. The magnetic bands are coloured blue, while the green and orange bands correspond to the time-reversal symmetric insulator bands. The insulator is in its trivial phase. The magnetic field strength is $b_z = 0.1$.

The investigated geometries are a bulk geometry, with periodic boundary conditions in both x- and y-directions, and a strip geometry, with periodic boundary conditions only in the y-direction and 20 lattice points wide in the x-direction.

3.1.2 Phase transition and Chern number

It is worth going into the details in the following. In the case of no inter-layer coupling, as expected, the topological phase of the TI layer has a well-defined intrinsic bulk band gap traversed by propagating edge states. Depending on the chemical potential, the energy levels of the magnetic layer might cross this band gap. However, since the two layers are not connected, the magnetic bands actually do not hybridise the TI bands and cannot destroy the intrinsic band gap. When the coupling is turned on, the magnetic bands are no longer isolated, and thus they are able to hybridise the TI bands, breaking the TRS. Nevertheless, the interaction of the magnetic-type bands and the insulator-type bands forms a different gap. This secondary gap is formed in the intrinsic gap of the non-coupled insulator, due to the repulsion between the magnetic states and the TI valence states (which together constitute the lower one-third of the fully hybridized band structure). The size of the newly opened gap depends largely on the strength of the magnetic field. A larger the magnetic filed requires a stronger coupling to open the secondary gap (see Figure 3). One might also observe that the edge states remained mostly intact, nevertheless, the topological protection is naturally broken due to the coupled magnetic layer.

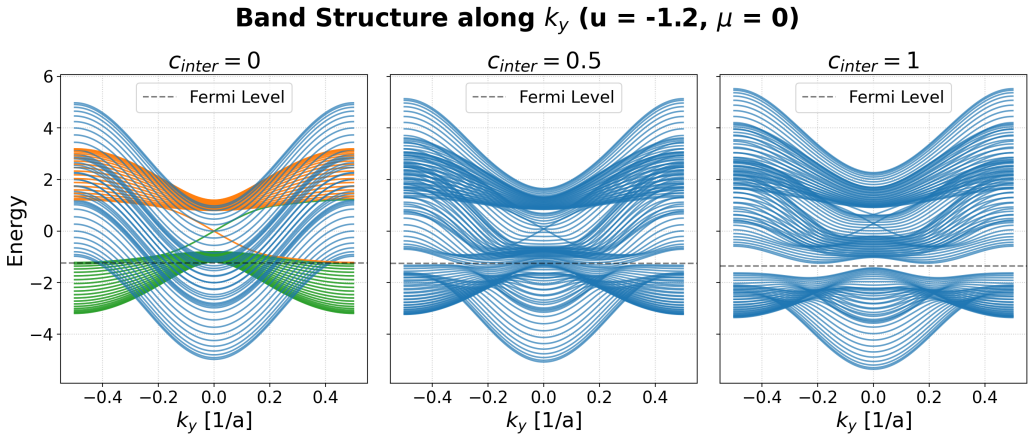


Figure 3: Band structure of the system along the k_y axis. In a stronger magnetic setting ($b_z = 1$), larger coupling is required to reopen a gap.

To further illustrate the phase transition of the material, the Chern number of the lower one-third of the bands was calculated using the `Z2Pack` package. Naturally, any change in the Chern number is expected only if the decoupled insulator is in its topological phase. Also, the Chern number requires the system to have a band gap in the bulk band structure. Therefore,

some combinations of parameters were selected to illustrate the existence or non-existence of the gap in Figure 4.

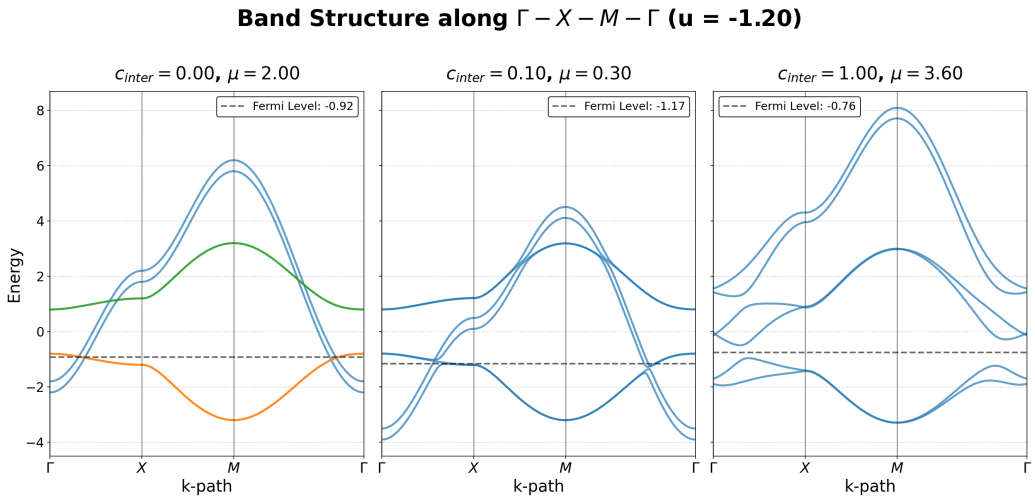


Figure 4: Band structure of the bulk system along the high symmetry points.

The phase transition is presented as a function of the chemical potential and the coupling between the layers, as shown in Figure 5. In all of the displayed cases, a similarly shaped new phase region is apparent. The width of the region is affected by the magnetic variable, whereas the Dirac mass shifts the position of the region on the μ axis.

3.1.3 Magnetic Moment

In the presence of strong spin-orbit coupling, the total magnetic moment consists of both a spin and an orbital contribution. Although the orbital magnetic moment can be significant in such relativistic cases, the isolated spin magnetic moment remains a relatively good parameter for characterising the macroscopic magnetic phase of the system. For the bulk system, the macroscopic spin magnetic moment is the expectation value of the spin operator over the occupied Bloch states. The total bulk magnetization along the z-axis, M_z, is obtained by integrating over the Brillouin zone (BZ) [16]:

$$M_z = \frac{1}{A_{\mathrm{BZ}}} \int_{\mathrm{BZ}} \mathrm{d}\vec{k} \sum_{n}^{E_{n,\vec{k}} < E_F} \langle \Psi_{n,\vec{k}} | \hat{m}_z | \Psi_{n,\vec{k}} \rangle \ , \tag{68}$$

where $|\Psi_{n,\vec{k}}\rangle$ are the eigenstates of the Bloch Hamiltonian with energy eigenvalues $E_{n,\vec{k}}$, and the summation runs over the occupied bands below the

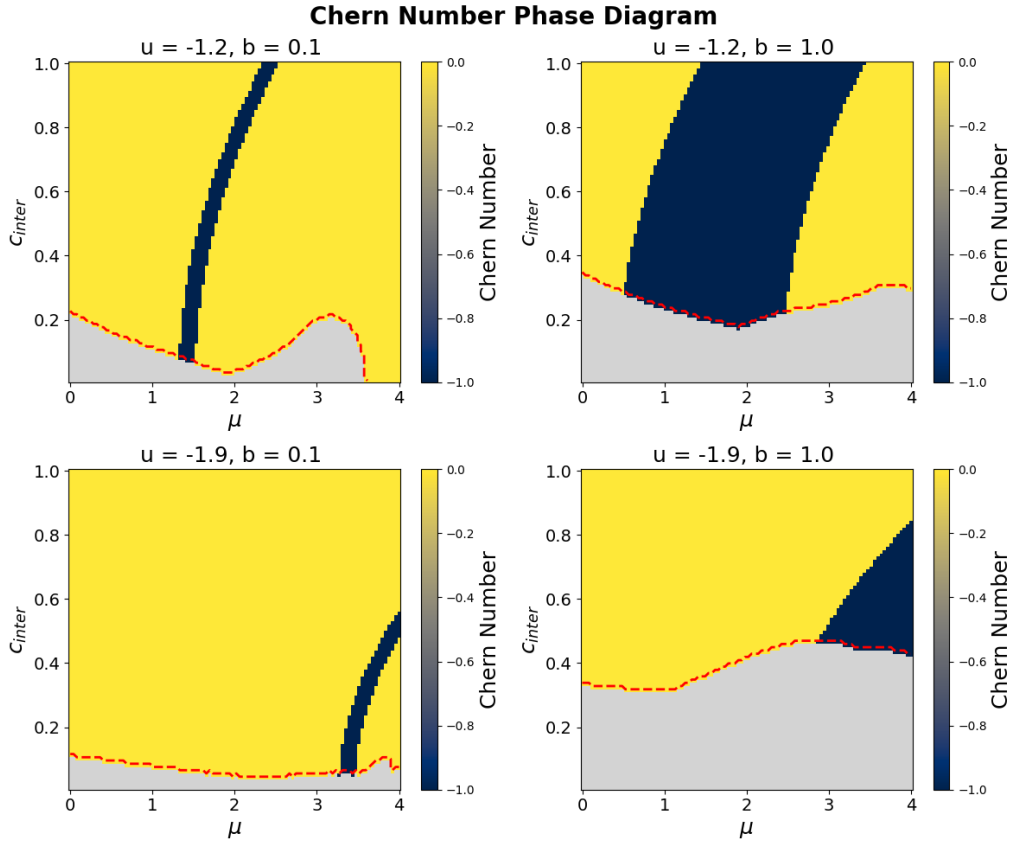


Figure 5: Phase transition of the bulk material as a function of the chemical potential and the inter-layer coupling. The Chern number was calculated on the lower one-third of the bands, because if μ is large, this setting would be a half-filled BHZ insulator. In that case, the expected Chern number is 0. However, the Chern number is well-defined only if at that Fermi level, dictated by the occupied lower one-third of the bands, the system is an insulator. Therefore, prior to the Chern number calculations, it had to be checked whether the system had a band gap at the Fermi level. The grey zone indicates the gapless, conducting regime.

Fermi energy (E_F). The operator $\hat{m}_z = \mathbb{I}_{\text{orb}} \otimes -\hat{s}_z$ is the z-projection of the magnetic moment in the spin space. For the discrete case the M_z is

$$M_z \approx \frac{1}{N_k} \sum_{\vec{k}} \sum_{n}^{E_{n,\vec{k}} < E_F} \langle \Psi_{n,\vec{k}} | \hat{m}_z | \Psi_{n,\vec{k}} \rangle \ . \tag{69}$$

Naturally, the Fermi level plays a crucial role; however, the band filling is tuned by adjusting the chemical potential shift instead.

Figure 6 illustrates how the tuning of the chemical potential, along with the magnetic exchange parameter, influences the resulting spin magnetic moment. Because the Fermi level is set to make the lower one third of bands filled, the chemical potential shift of the magnetic layer must be negative and sufficiently large to separate its states from the insulator bands and to make the magnetic bands occupied, instead of the insulator bands. As the bands are separated in their energy, the occupied bands are becoming increasingly magnetic, thus the magnetic moment increases.

Additionally, a large magnetic exchange parameter is required to reach a value of $1\,\mu_B$, which is the theoretical maximum spin magnetic moment for a single, fully polarised spin-$\frac{1}{2}$ band.

3.2 Extraction of Magnetic Exchange Parameters

After the real-space Hamiltonian is created, we can turn our attention to extracting the magnetic exchange interactions of the extended Heisenberg model. For this purpose, the `grogupy` package was used [17].

The package relies on the Liechtenstein-Katsnelson-Antropov-Gubanov (LKAG) torque method, utilising Green's functions to calculate the relevant exchange interaction tensors [18, 19]. In the context of our constructed toy model, a "localised magnetic moment" corresponds to the effective magnetic orbital residing at a specific lattice site, which is subjected to the local magnetic exchange field ($\vec{b}$). By treating infinitesimal spatial rotations of these localised moments as perturbations, the exchange interactions can be computed. In the classic collinear limit without spin-orbit coupling, the isotropic exchange interaction (J_{ij}) between site i and site j is computed directly from the real-space Green's functions in the energy domain [19]:

$$J_{ij} = \frac{1}{4\pi} \text{Im} \int_{-\infty}^{E_F} \mathrm{d}E \, \text{Tr} \left[\hat{\Delta}_i \, \hat{G}^{\uparrow}_{ij}(E) \, \hat{\Delta}_j \, \hat{G}^{\downarrow}_{ji}(E) \right] \ , \tag{70}$$

where $\hat{G}^{\uparrow/\downarrow}_{ij}(E)$ are the spin-projected retarded Green's functions propagating electrons between the sites. The term $\hat{\Delta}_i$ represents the local magnetic exchange splitting matrix. In our specific model, this splitting is driven entirely

by the applied exchange field, meaning $\hat{\Delta}_i$ is directly proportional to $\vec{b} \cdot \hat{\boldsymbol{s}}$, and is entirely independent of the global chemical potential shift ($\mu \cdot \hat{s}_0$).

However, because the underlying tight-binding grid of the topological insulator incorporates strong spin-orbit coupling, spin is no longer a conserved quantity. Consequently, the simple spin-collinear projection in (70) is insufficient to describe the full physics of the system. To capture the complete relativistic magnetic interactions, the LKAG formalism must be generalised. The `grogupy` package achieves this by evaluating the full vector trace in the combined spin-orbital space, utilising fully relativistic, non-diagonal Green's function matrices ($\hat{G}_{ij}(E)$) [20]. By calculating the variations of the band energy with respect to independent local rotations of the magnetic moments along the spatial axes ($\alpha, \beta \in \{x, y, z\}$), the elements of the generalised 3×3 exchange tensor $\bar{\bar{\mathcal{J}}}_{ij}$ are given by:

$$\mathcal{J}_{ij}^{\alpha\beta} = \frac{1}{4\pi} \mathrm{Im} \int_{-\infty}^{E_F} \mathrm{d}E \, \mathrm{Tr} \left[\hat{\Delta}_i \hat{s}_\alpha \hat{G}_{ij}(E) \hat{\Delta}_j \hat{s}_\beta \hat{G}_{ji}(E) \right] , \tag{71}$$

where and the trace is computed over the fully coupled spin-orbital space.

Based on the construction of the real-space Hamiltonian, specific symmetries are expected to govern the DMI vectors. First, the DMI is constrained by the spin-orbit coupling. Since the x-directional hopping operator ($\hat{T}_x$) has only one spin-orbit coupling term ($\hat{s}_z$), this polarises any resulting DMI vectors purely along the z-axis. This symmetry restriction justifies our earlier choice to apply the magnetic exchange field solely in the z-direction ($\vec{b} = (0, 0, b_z)$), as it aligns with the only non-vanishing components of the Dzyaloshinskii-Moriya interaction. Second, the emergence of the DMI requires simultaneously broken inversion symmetry and SOC. While $\hat{T}_x$ possesses SOC terms, the hopping operator along the y-axis has none (since it only contains terms proportional to the identity operator). Consequently, electron transport along this axis is spin-conserving, and all the DMI magnitudes for any pair of neighbours located exactly along the y-axis must vanish. The spatial inversion symmetry of the system is broken by $\hat{U}$. Specifically, it arises from the combination of the symmetric inter-layer couplings and the opposite parities of the mass parameters. To demonstrate this, define the spatial inversion operator as

$$\hat{\mathcal{I}} = \begin{bmatrix} \hat{s}_0 & 0 & 0 \\ 0 & -\hat{s}_0 & 0 \\ 0 & 0 & P_m \hat{s}_0 \end{bmatrix} , \tag{72}$$

where $P_m \in \{-1, +1\}$ is the parity of the magnetic orbitals. Applying this

on $\hat{U}$ yields

$$\hat{\mathcal{I}}\hat{U}\hat{\mathcal{I}}^{-1} = \begin{bmatrix} u\hat{s}_0 & 0 & P_m c_{\text{inter}}\hat{s}_0 \\ 0 & -u\hat{s}_0 & -P_m c_{\text{inter}}\hat{s}_0 \\ P_m c_{\text{inter}}\hat{s}_0 & -P_m c_{\text{inter}}\hat{s}_0 & \vec{b}\cdot\hat{\boldsymbol{s}} + \mu\hat{s}_0 \end{bmatrix} . \tag{73}$$

From this, it is clear, that P_m should be 0 for $\hat{U}$ to remain invariant, thus the system breaks the inversion symmetry.

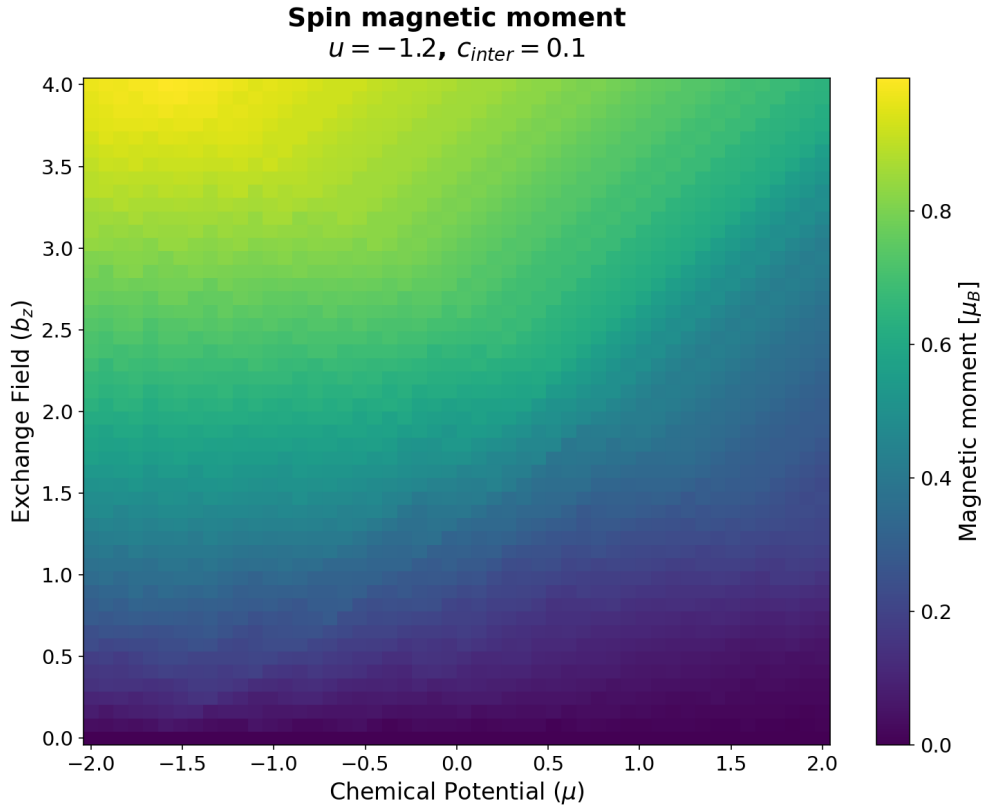


Figure 6: The change of the macroscopic bulk spin magnetic moment is displayed as a function of the chemical potential and the magnetic exchange field. The other system parameters are kept at $u = -1.2$ and $c_{inter} = 0.1$.

4 Results

4.1 Convergence calculations

To carry out the integration in Eq. (70), one must first calculate the Green's functions and then set the numerical bounds of the integration and the number of integration points. To do so, the lower limit can be any value that is below the lowest energy band; in the following results, it is set to -10. The upper bound is the Fermi level, which is set in each case so the number of occupied bands is one-third of all bands (representing a half-filled insulator layer if the magnetic bands are pushed in a higher energy level by the μ parameter). The convergence calculations connected to the integral grid size of the Green's function k-space and is shown in Figure 7. To evaluate the convergence of the integrals, the convergence of the isotropic exchange interaction J_{iso} is used as a measure. Since the absolute magnitude of J_{iso} decays rapidly with increasing interatomic distance, direct comparisons of the raw coupling values are impractical. To overcome this difficulty, the calculated values for each unique atomic distance are normalised by their respective mean across the entire parameter sweep. To extract the global behaviour, one must then aggregate the normalised data across all distances, then the median is calculated at each computational step. As the results of the convergence calculations show, the step size of the energy contour will be $n_{step} = 20$ and the k-grid size will be $(200 \times 200 \times 1)$. With the previous results established, we can turn our attention back to the Fermi level calculations. The Fermi level will be calculated with the built-in self-consistent function of the `sisl` package on a $(200 \times 200 \times 1)$ k-grid.

4.2 Bulk case

The convergence calculations were followed by an investigation of the bulk geometry. The purpose of the simulations was to map the isotropic magnetic interactions and the DMI terms across the parameter space (Table 1). Throughout all subsequent calculations, the exchange tensor components were extracted as a function of the neighbour distance. To visualise these tensor components, additional scalar quantities had to be derived. The maximum eigenvalue of the traceless symmetric and antisymmetric parts of the exchange tensor was used as a norm. Because the antisymmetric tensor is expected to have only one independent component, this introduced norm is equivalent to the maximum component of the corresponding DM vector; thus, these definitions will be used interchangeably.

Figure 8 presents the results of a typical calculation. Looking at the fig-

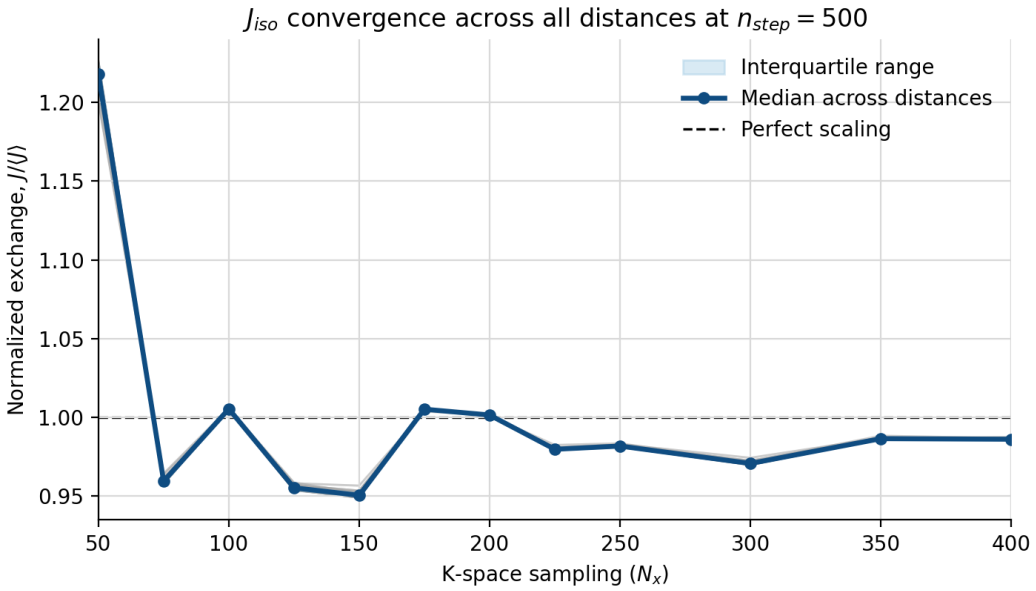


Figure 7: The calculated median of the normalised isotropic exchange terms at each k-grid size ($N_x \times N_y \times 1$).

ure, it is visible that there are no relativistic contributions in the case of a completely decoupled system, as expected. Moreover, the traceless symmetric part of the exchange tensor is orders of magnitude smaller in all cases than the other components; therefore, it will be excluded from further discussion. Additionally, scaling the magnetic parameter changed the overall amplitude of the isotropic terms, while the shape of their spatial distribution remained very similar. The same cannot be said with such confidence about the DMI terms. Changes in the system parameters resulted in relatively different magnitudes and overall shape. Nonetheless, to quantitatively evaluate these observation, other comparison techniques were required.

The most comprehensive visualisations display the n-th neighbour interaction terms as a function of specific system parameters. Among these, the chemical potential shift of the magnetic layer (μ) emerged as one of the most impactful. However, before one can compare the results directly, due to the large magnitude differences generated by the various parameter combinations, the terms must be normalised. Consequently, each data set for a particular parameter setting was normalised by the maximum of its absolute value. Finally, since there are four n-th neighbours for a given distance on the square lattice, the mean of the interaction terms is plotted. The following figures (Figure 9 and Figure 10) depict the typical results of such normalised data points.

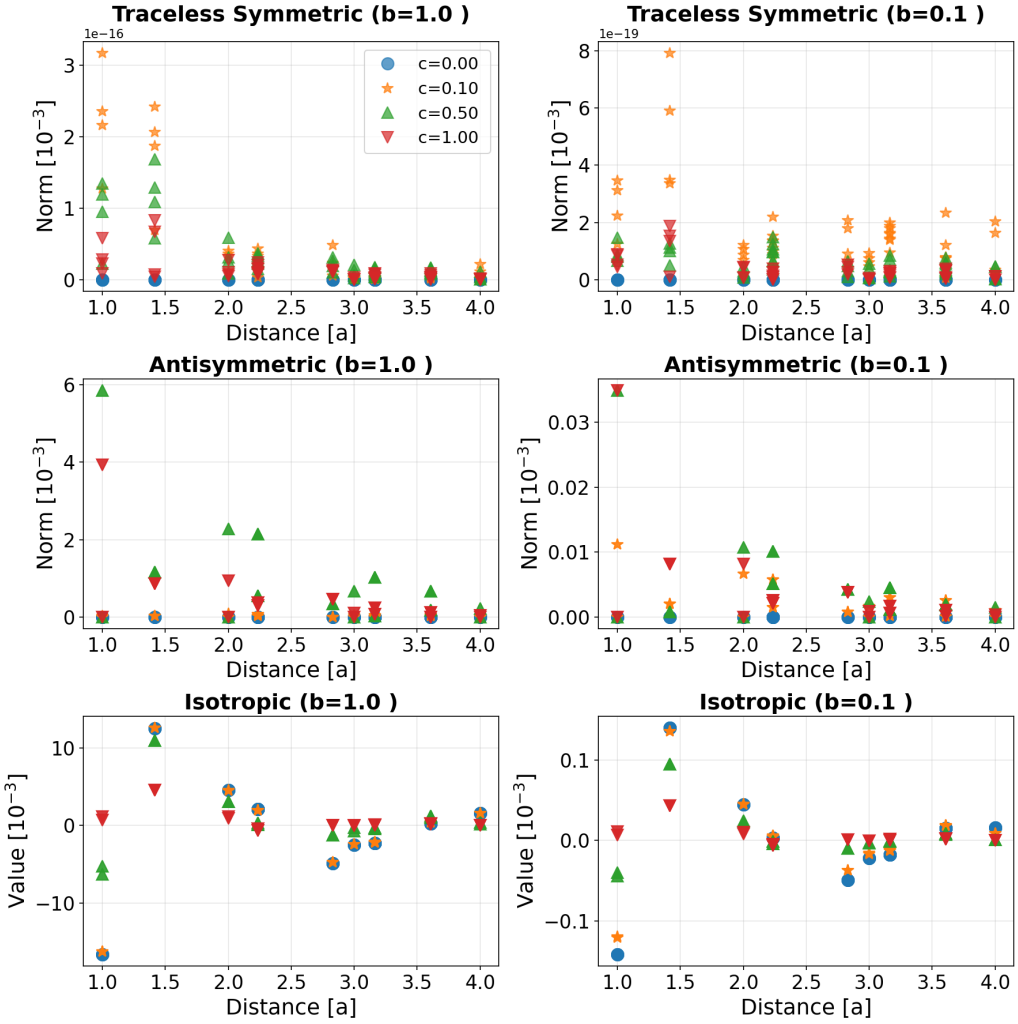


Figure 8: The different metrics of the exchange tensor is presented as a function of atomic distance. The system parameters were $u = -1.2$, $\mu = 1.0$, and all the others are indicated in the figure.

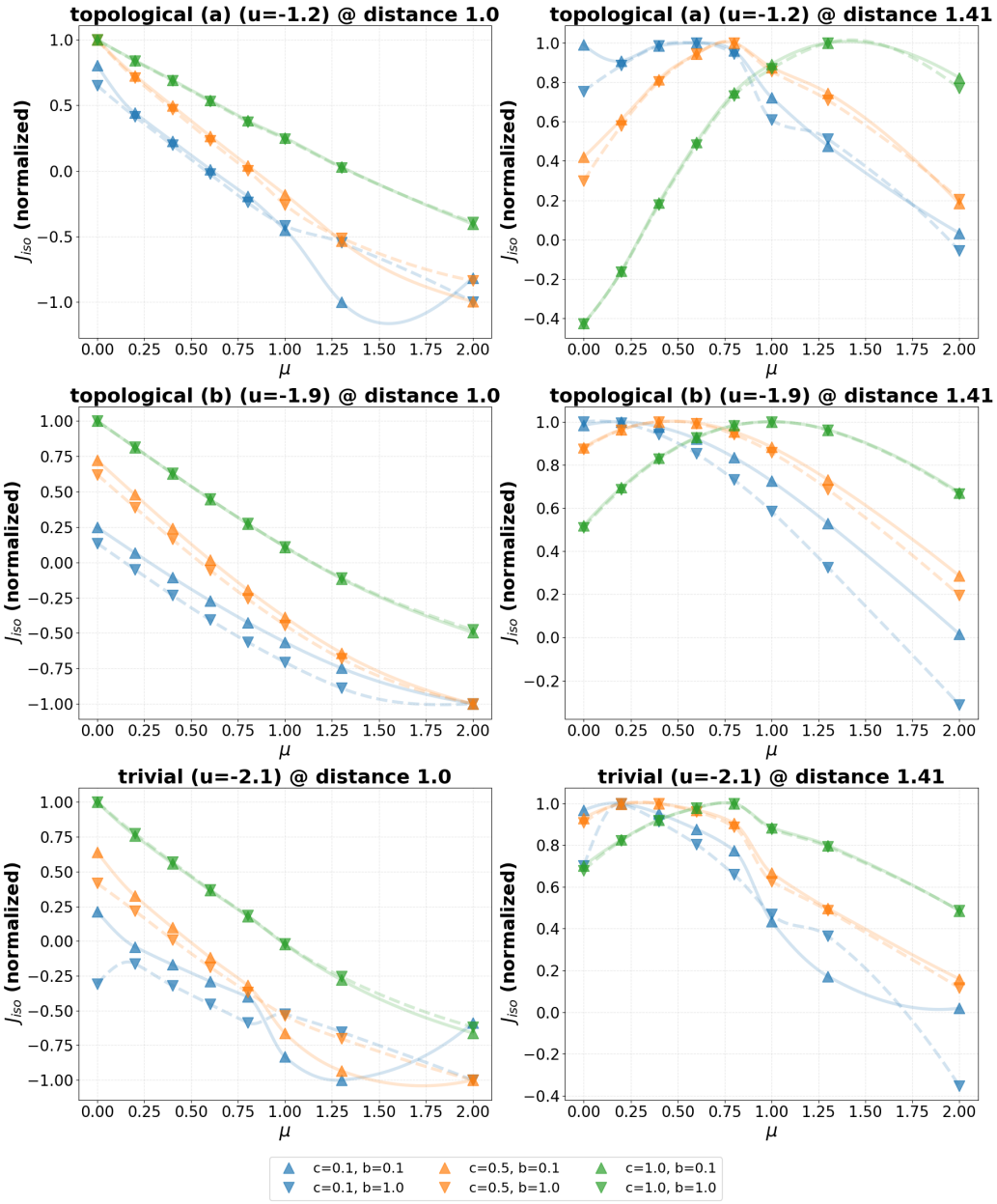


Figure 9: First and second-neighbour isotropic interaction terms of a bulk system as a function of the chemical potential shift of the magnetic layer (μ). The dotted and continuous lines are interpolated points. All the displayed quantities are averages over neighbours of the same distance.

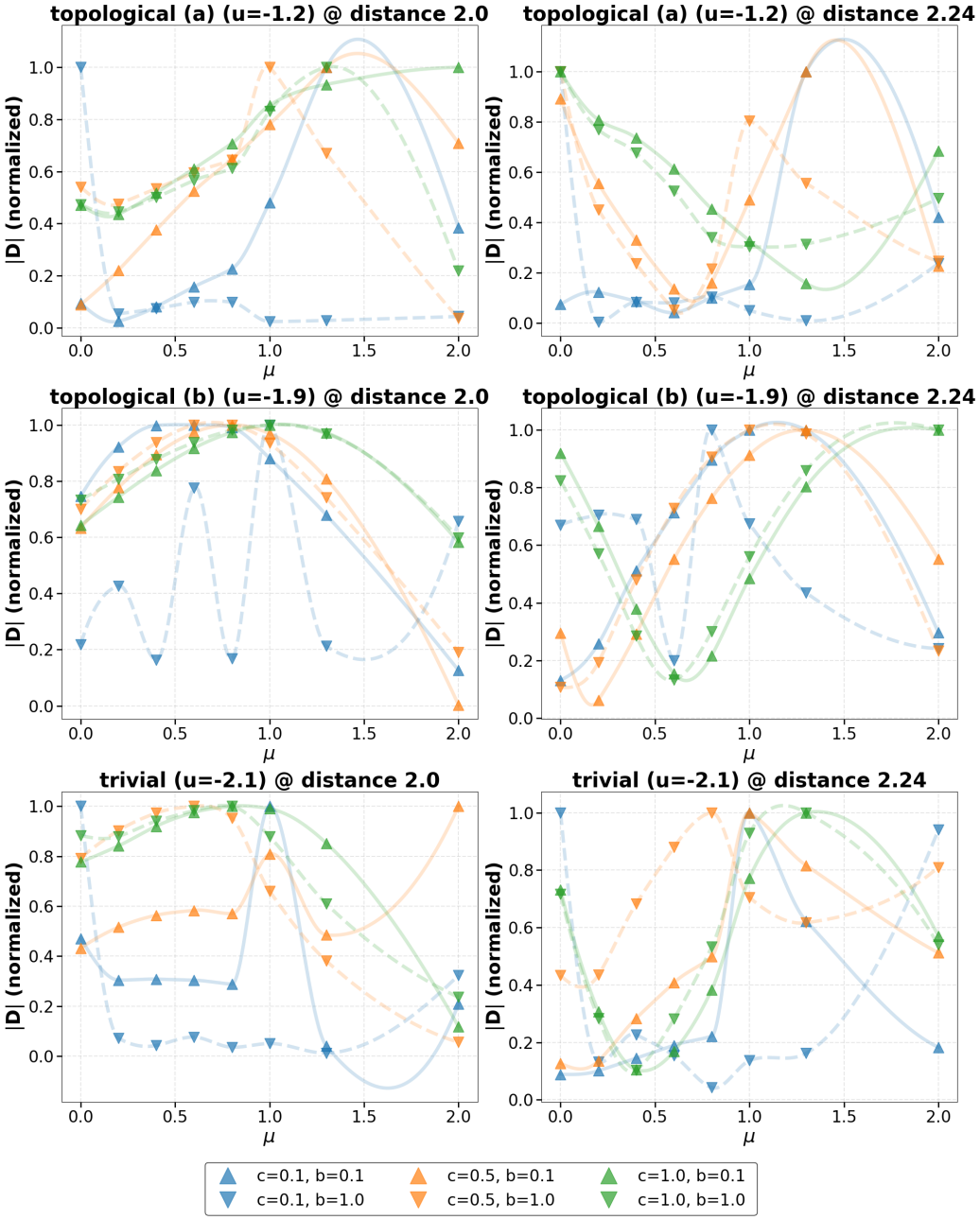


Figure 10: Third and fourth-neighbour DMI terms of a bulk system as a function of the chemical potential shift of the magnetic layer (μ). The dotted and continuous lines are interpolated points. All the displayed quantities are averages over neighbours of the same distance.

After thorough analysis of the resulting figures, several observations were made. First, just as previously discussed, the DMI terms are more sensitive to the parameter configuration than the isotropic ones. Any changes in either the magnetic or the coupling parameters resulted in a drasticly different behaviour. Secondly, the magnetic parameter has a minimal effect on the qualitative shape of the curves, acting primarily to scale the overall amplitude of the magnetic interaction, unless the interlayer coupling (c_{inter}) is very small. This can be confirmed by comparing the differently shaped markers for each individual colour. At low inter-layer coupling, the corresponding curves for different b cases do not perfectly overlap, which means that the magnetic parameter not only affected the exchange term magnitudes, but introduced non-trivial shape effects as well. Finally, tuning the Dirac mass parameter does not seem to alter the overall shape of the curves in a drastic way. In all cases, the first- and second-neighbour isotropic interaction terms exhibit a sign change as a function of μ, indicating a transition from a ferromagnetic to an antiferromagnetic state. Comparing these zero-crossings across different values of u, it is clear that the transitions happen at shifted values of μ. This structural similarity, alongside the shifted zero-crossings, suggests that modifying u produces an effect analogous to shifting the chemical potential μ. This behaviour can be understood by considering the Fermi level, which is consistently set to fill the lower one-third of the bands. As the magnitude of u increases, the spin-down bands are shifted to lower energy levels, which decreases the hybridisation between the different band types. This mechanism effectively mirrors an increase in the chemical potential shift, which similarly reduces the hybridisation between the magnetic bands and the spin-down insulator bands. However, because increasing u does not produce a uniform energy separation between the spin-up and spin-down bands across all $\vec{k}$-points, the effects of u and μ are not equivalent. This limitation is demonstrated by the altered interaction curve shapes in the $u = -1.2$ case.

Another key aspect of the investigation was the spatial isotropy of the exchange terms. In combinations of specific parameters, there were distinguishable differences in the isotropic exchange terms of n-th neighbours along the x- and y-axes, as seen in Figure 11. Once again, the key factor governing this phenomenon was the interlayer coupling strength, strong coupling restored spatial isotropy to the system. Additionally, the transition of u from the trivial regime to the topological regime did not alter this general isotropic behaviour. Beyond the isotropic Heisenberg exchange terms, the Dzyaloshinskii-Moriya interaction (DMI) terms also exhibited interesting symmetries. First, because the exchange field was applied only along the z-axis (b_z), the resulting DM vectors were expected to possess only z-components. This assumption was confirmed in all setups evaluated. Second,

it was observed that the magnitude of the DM interaction vectors vanished for all pairs of lattice points located exactly along the y-axis.

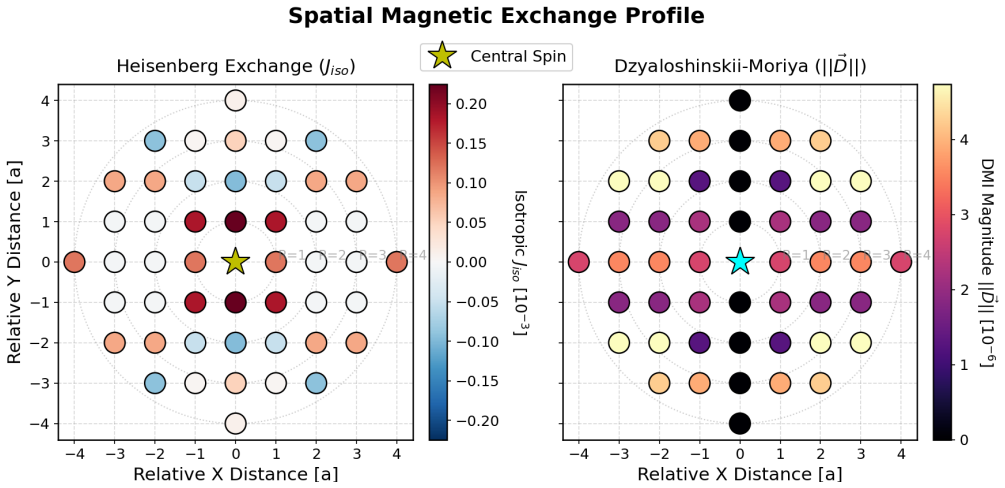


Figure 11: Spatial magnetic exchange profile of a system with parameters $u = -1.2$, $\mu = 0$, $c_{inter} = 0.1$, and $b = 0.1$.

The directional dependencies of both the isotropic and DM terms arise from the underlying anisotropy built into the bulk Hamiltonian (61). Taking a closer look, the hopping amplitudes in the x-direction are purely imaginary, while those in the y-direction are real:

$$\hat{H}_x(\mathbf{k}) = \hat{s}_0 \otimes \left(\frac{\mathrm{e}^{\mathrm{i}k_x} + \mathrm{e}^{-\mathrm{i}k_x}}{2} \right) \hat{\sigma}_z + \hat{s}_z \otimes \left(\frac{\mathrm{e}^{\mathrm{i}k_x} - \mathrm{e}^{-\mathrm{i}k_x}}{2\mathrm{i}} \right) \hat{\sigma}_x \,, \tag{74}$$

$$\hat{H}_y(\mathbf{k}) = \hat{s}_0 \otimes \left(\frac{\mathrm{e}^{\mathrm{i}k_y} + \mathrm{e}^{-\mathrm{i}k_y}}{2} \hat{\sigma}_z + \frac{\mathrm{e}^{\mathrm{i}k_y} - \mathrm{e}^{-\mathrm{i}k_y}}{2\mathrm{i}} \hat{\sigma}_y \right) \,, \tag{75}$$

where $\hat{\sigma}_x/\mathrm{i}$ is purely imaginary, while $\hat{\sigma}_y/\mathrm{i}$ is real. Applying a unitary rotation in either the spin or the orbital space preserves the time-reversal symmetry of the system and yields identical energy eigenstates. A $\pi/4$ rotation around the x-axis in the spin space renders all hopping amplitudes real (via the unitary operator $\hat{U} = \mathrm{e}^{-\mathrm{i}\pi/4 \cdot \hat{s}_x} \otimes \hat{\sigma}_0$):

$$\hat{U}\hat{H}_x(\mathbf{k})\hat{U}^\dagger = \hat{s}_0 \otimes \left(\frac{\mathrm{e}^{\mathrm{i}k_x} + \mathrm{e}^{-\mathrm{i}k_x}}{2} \right) \hat{\sigma}_z - \hat{s}_y \otimes \left(\frac{\mathrm{e}^{\mathrm{i}k_x} - \mathrm{e}^{-\mathrm{i}k_x}}{2\mathrm{i}} \right) \hat{\sigma}_x \,, \tag{76}$$

with the other parts remaining unchanged. Additionally, making all amplitudes purely imaginary is achievable through a combined rotation in both orbital and spin space (the spin rotation is only necessary if the intra-layer

coupling is non-zero,) using the operator $\hat{U}' = \mathrm{e}^{-\mathrm{i}\pi/4\cdot\hat{s}_z} \otimes \mathrm{e}^{-\mathrm{i}\pi/4\cdot\hat{\sigma}_x}$:

$$\hat{U}'\hat{H}_x(\mathbf{k})\hat{U}'^{\dagger} = \hat{s}_0 \otimes \left(\frac{\mathrm{e}^{\mathrm{i}k_x} + \mathrm{e}^{-\mathrm{i}k_x}}{2}\right)(-\hat{\sigma}_y) + \hat{s}_z \otimes \left(\frac{\mathrm{e}^{\mathrm{i}k_x} - \mathrm{e}^{-\mathrm{i}k_x}}{2\mathrm{i}}\right)\hat{\sigma}_x\,, \quad (77)$$

$$\hat{U}'\hat{H}_y(\mathbf{k})\hat{U}'^{\dagger} = \hat{s}_0 \otimes \left[\frac{\mathrm{e}^{\mathrm{i}k_y} + \mathrm{e}^{-\mathrm{i}k_y}}{2}(-\hat{\sigma}_y) + \frac{\mathrm{e}^{\mathrm{i}k_y} - \mathrm{e}^{-\mathrm{i}k_y}}{2\mathrm{i}}\hat{\sigma}_z\right]. \quad (78)$$

The newly extracted exchange terms using the rotated Hamiltonians are presented in Figure 12. Transforming the Hamiltonian successfully eliminated the directional dependence of the isotropic exchange terms. However, the DMI terms remained direction-dependent, confirming theoretical expectations, since the y-axis hopping term still lacks intrinsic spin-orbit coupling.

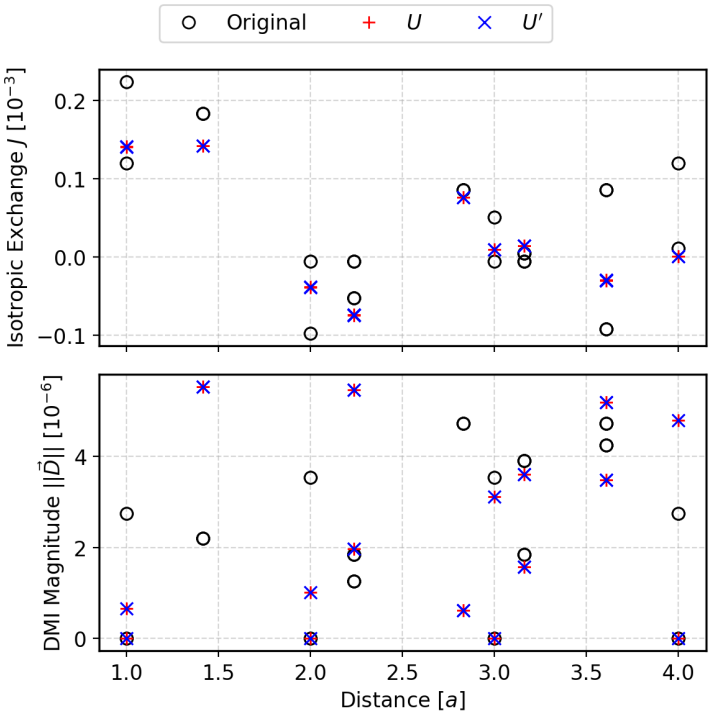


Figure 12: The calculated exchange terms comparing the rotated Hamiltonians to the original.

4.3 Strip geometry

Following the bulk analysis, attention was shifted to the strip geometry. Because the normalised first- and second-neighbour isotropic exchange term

analysis proved highly effective for understanding the influence of the system parameters, a similar methodology was applied to the strip configuration. The corresponding results are presented in Figure 13. Similar to the bulk case, we can observe that varying the magnitude of b affects the overall amplitude of the exchange terms. Surprisingly, unlike in the bulk system, the low inter-layer coupling did not alter this scaling effect of b. Nonetheless, the relationship between the μ and u parameters remains similar to the one seen in the bulk material. Specifically, the system goes through a phase change from ferromagnetic to antiferromagnetic, the zero-crossing of which is governed by the combination of μ and u.

Compared to the bulk case, evaluating the spatial distribution of the n-th neighbour interaction constants requires a different methodology, since the physical position of the lattice point within the strip might significantly alter its magnetic behaviour.

To demonstrate the effect of the lattice position within the strip geometry, the nearest-neighbour isotropic exchange terms are displayed for each lattice site in Figure 14. For each site, three bars are plotted: two represent the exchange interactions with the left and right neighbours along the x-direction, while the third represents the interaction along the y-direction. A single bar is sufficient for the y-direction because the up and down neighbours exhibit symmetrical behaviour. The presence of the physical edge causes significant deviations in these nearest-neighbour terms. For example, depending on the system configuration, the interactions near the boundary can experience a massive change in magnitude, or even a complete sign flip where otherwise ferromagnetic terms are replaced by antiferromagnetic ones. These edge effects are clearly visible across all parameter combinations in both the topological and trivial phases of the insulator. Therefore, they are not rooted in the presence of a topological edge state.

4.3.1 Edge effects

In the previous sections, the effects of parameter adjustments and lattice positions on the exchange terms have been explored. Based on these results, it is clear that these variables influence the overall magnetic behaviour of the material. To better understand the relationship between spatial proximity to the edge and the onset of these magnetic behaviours, a comparative study is necessary. The existence of the edge effects is established; however, due to their extremely high dependence on the parameter configuration, the emerging effects can only be compared to their corresponding bulk counterpart. This makes direct comparison across different configurations challenging, highlighting the need for a normalised measure of the edge effects.

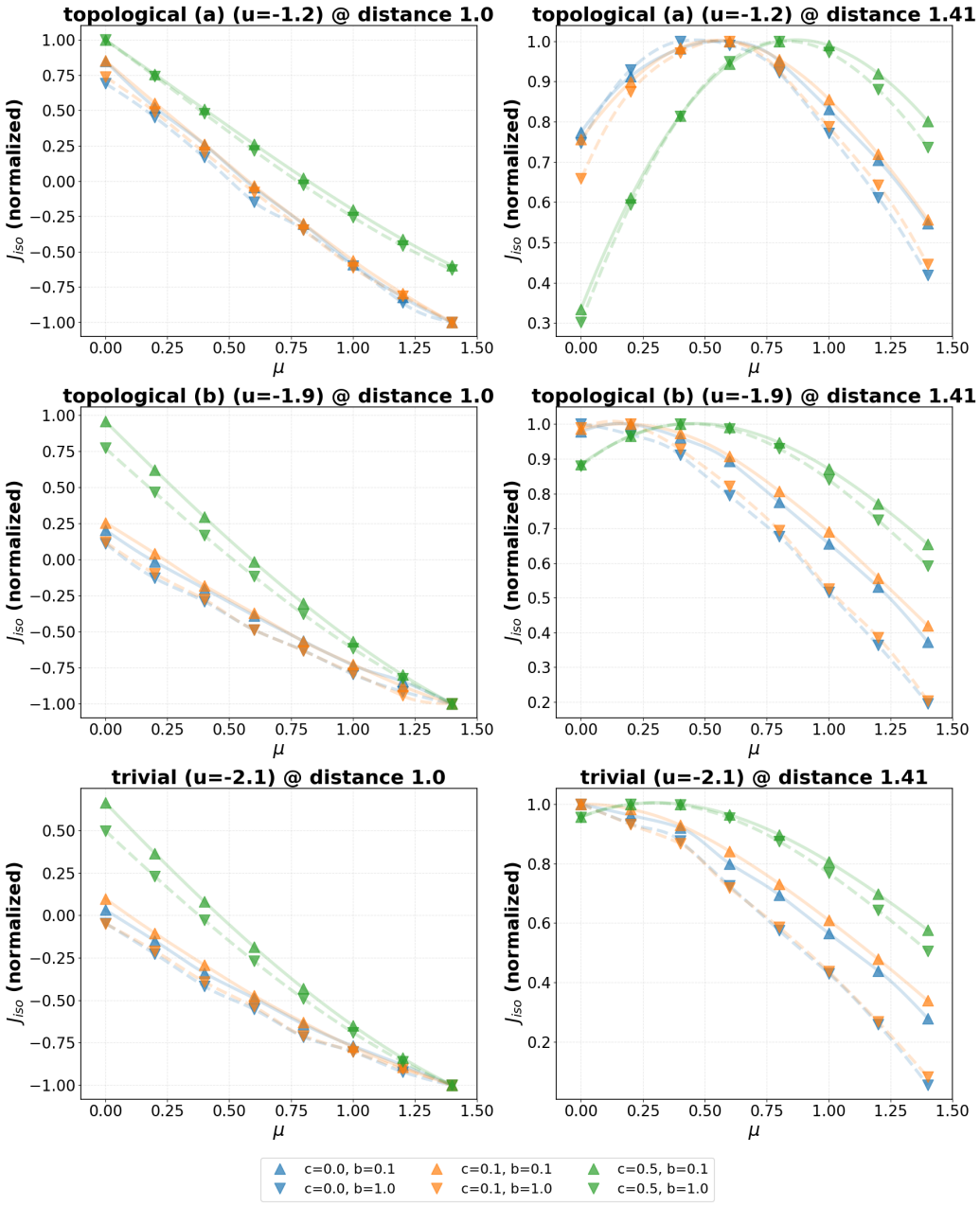
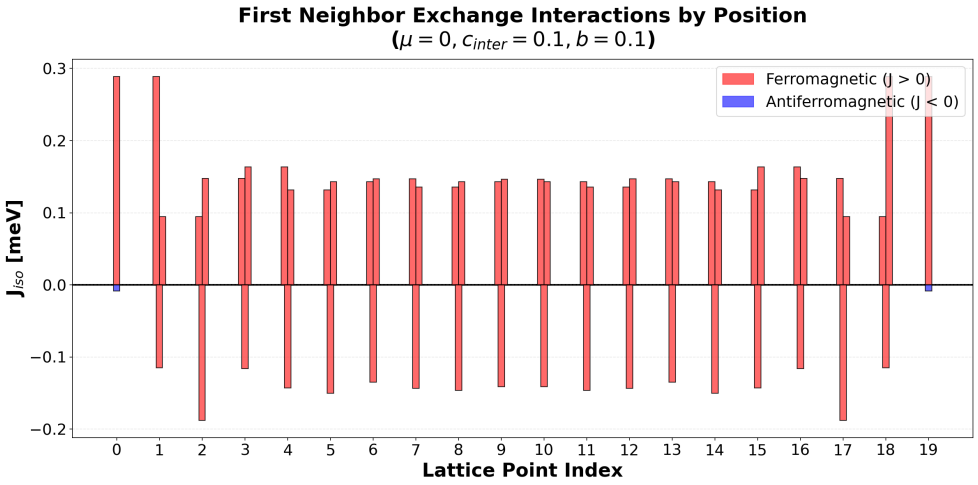
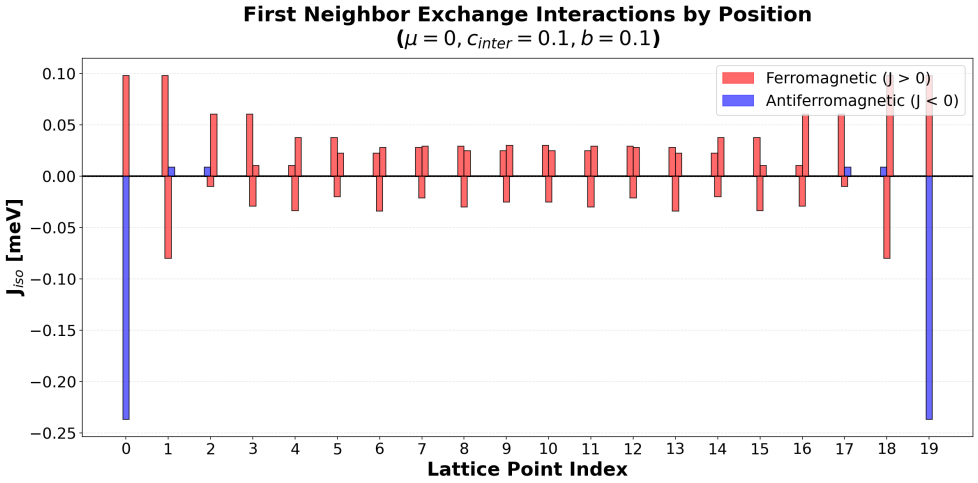


Figure 13: First and second-neighbour isotropic interaction terms of a strip system as a function of the chemical potential shift of the magnetic layer (μ). The dotted and continuous lines are interpolated points. All the displayed quantities are averages over neighbours of the same distance.



(a) Insulator layer topological phase



(b) Insulator layer trivial phase

Figure 14: Spatial distribution of the first-neighbour isotropic terms demonstrating edge effects. For each lattice point the first-neighbour isotropic terms are displayed, by displaying the magnitude of the terms on the y-axis. The corresponding terms for the neighbours in the y-direction are shown by all negative values, whilst the positive values are assigned to the x-direction neighbours. The signs of the exchange terms are represented by the colour of the bars.

Consequently, all resulting exchange terms for the various parameter combinations and strip lattice positions are evaluated against their corresponding infinite bulk terms. Because boundary truncation in the strip geometry reduces the total number of exchange terms relative to the bulk, only the overlapping terms are taken into consideration. For each lattice point i and parameter combination $\{p\}$, the corresponding isotropic magnetic exchange terms are gathered into a vector $\vec{\alpha}_i^{\{p\}}$, which is then compared against the identically constructed bulk vector $\vec{\beta}^{\{p\}}$. For each lattice index, the cosine similarity between the two vectors is calculated:

$$S_i^{\{p\}} = \frac{\langle \vec{\alpha}_i^{\{p\}}, \vec{\beta}^{\{p\}} \rangle}{||\vec{\alpha}_i^{\{p\}}|| \cdot ||\vec{\beta}^{\{p\}}||} \,.$$

This similarity measure is highly suitable for our case, since it disregards the absolute magnitudes of the vectors, and only measures the qualitative shape similarity. It yields a maximum value of one if the vectors point in the exact same direction (i.e., one is a positive scalar multiple of the other), evaluates to zero if they are strictly orthogonal, and evaluates to negative one if they point in opposite direction. The resulting similarities are presented in Figure 15.

Echoing previous observations, weak coupling regimes display different decay profiles than strongly coupled systems. The curves corresponding to $u = -1.9$ differ from the rest of the dataset. For this specific mass parameter, the calculated similarities are consistently close to 1 across the entirety of the strip, indicating a suppression of spatial edge effects compared to the infinite bulk. In contrast, the other two mass cases exhibit a sharp initial rise in similarity from the physical boundaries, stabilising at a steady plateau in the interior. To confirm that the $u = -1.9$ case constitutes a statistical outlier, Principal Component Analysis (PCA) was performed on the gathered similarity profiles (see Figure 16). The resulting PCA projection successfully isolates the $u = -1.9$ configurations, confirming their outlier nature.

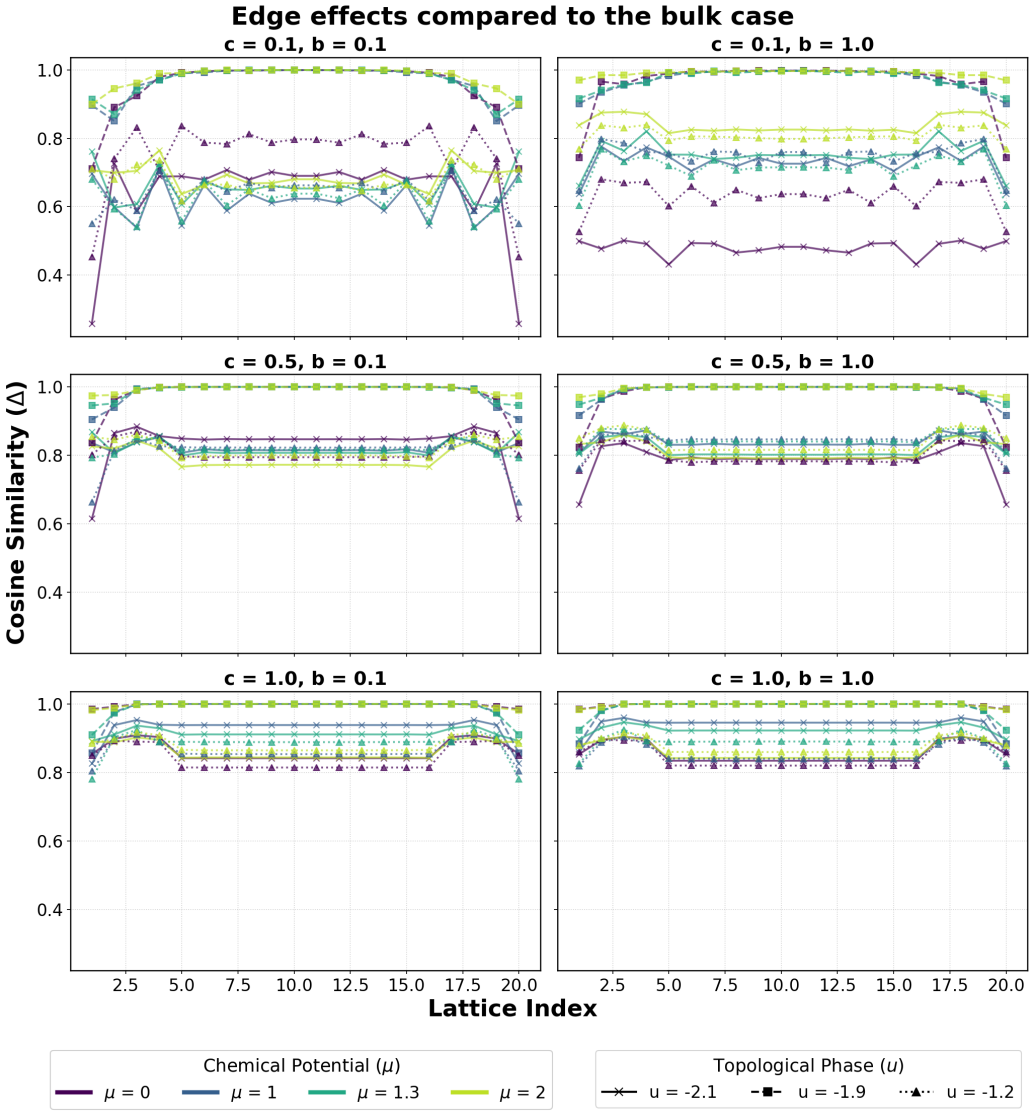


Figure 15: The calculated cosine similarities $S_i^{\{p\}}$ as a function of lattice indices, for each parameter combination. The similarities begin lower near the boundaries, indicating strong edge effects, then rapidly rise, then sink, and then plateau as the edge effects phase out in the bulk-like interior.

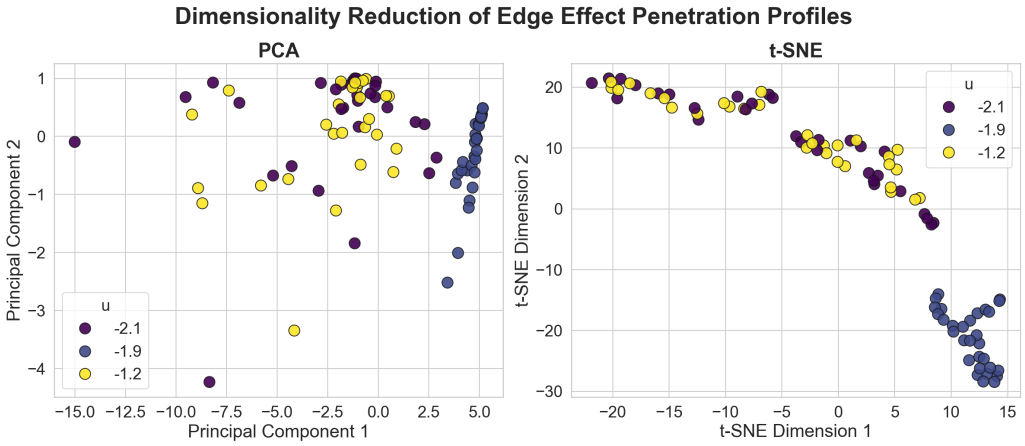


Figure 16: On the left, the PCA projection is displayed. The blue points, representing the $u = -1.9$ cases, form a separated cluster. On the right, the results of a t-distributed stochastic neighbour embedding (t-SNE) are displayed to further highlight the distinct grouping of these states.

5 Conclusion

In this thesis, the effects of proximity-induced magnetism on topological edge states were investigated through the construction and analysis of a discrete real-space tight-binding model. Based on the Bernevig-Hughes-Zhang (BHZ) formalism, the study modelled a two-dimensional topological insulator coupled to a localised magnetic layer. By introducing an effective magnetic exchange field within the heterostructure, the mechanisms of time-reversal and inversion symmetry breaking and their impact on the electronic band structure were mapped. A central achievement of this work is the characterisation of the topological phase transition driven by inter-layer coupling. It was demonstrated that upon sufficient hybridisation between the topological and magnetic orbitals, the intrinsic time-reversal-protected band gap closes, destroying the $\mathbb{Z}_2$ helical edge states. Crucially, the strong level repulsion between the magnetic states and the topological valence bands opens a secondary hybridisation gap. Within this newly formed gap, the system transitions into new phase characterised by a calculated Chern number of $C = 1$.

Beyond the electronic band structure, the magnetic properties of this heterostructure were quantified by extracting the isotropic Heisenberg exchange couplings and the Dzyaloshinskii-Moriya interaction vectors. These calculations were performed utilising a Green's function-based torque method. The extracted exchange interactions initially exhibited a highly distinct spatial anisotropy, concerning the isotropic exchange and the DMI, which vanished entirely along the y-axis. By applying unitary rotations in the spin and orbital spaces of the tight-binding Hamiltonian, it was shown that this directional dependence of the isotropic Heisenberg interaction is not a numerical artefact, but rather a consequence of symmetries of the underlying BHZ Hamiltonian. The directional dependence of the DMI was shown to be the result of the anisotropic spin-orbit coupling built into the Hamiltonian.

Through a comparative analysis of the bulk and strip geometries, it was established that the Dzyaloshinskii-Moriya interaction terms are significantly more sensitive to parameter variations than the isotropic exchange terms. For the isotropic interactions, the overall interaction amplitude is primarily scaled by the magnetic exchange field. In the bulk system, this straightforward scaling alters at weak inter-layer couplings, where the magnetic parameter introduces non-trivial deformations to the interaction profiles. Surprisingly, the strip geometry proved robust against this breakdown, preserving the simple scaling effect of b even at low coupling strengths.

Furthermore, the analysis demonstrated that the system undergoes a phase transition from a ferromagnetic to an antiferromagnetic state, driven

by the chemical potential shift (μ). The phase transition points are highly sensitive to the Dirac mass parameter (u), which shifts the zero-crossings of this transition. This produces an effect similar to a shift in μ. This can be explained by the fact that increasing the magnitude of u lowers the relevant energy levels and reduces hybridisation between the bands. However, the u-driven non-uniform energy separation across the Brillouin zone introduces other non-linear effects.

The transition from a periodic bulk lattice to a finite strip geometry revealed spatial edge effects in the magnetic exchange terms. Through cosine similarity calculations, the penetration depth of these edge effects from the physical boundaries into the bulk-like interior was quantified. The system exhibited an anomalous state at $u = -1.9$. For this specific Dirac mass, the spatial edge effects were almost entirely suppressed, resulting in bulk-like magnetic behaviour extending to the boundaries of the material. To classify this observation, principal component analysis was employed, which isolated the $u = -1.9$ configurations as a distinct cluster.

Although the developed tight-binding model successfully captures these topological phase transitions and magnetic exchange symmetries, certain limitations remain to be addressed in future work. To transition from qualitative mapping to quantitative material predictions, future iterations of this model should include experimentally fitted system parameters. Additionally, mapping the topological phase diagrams and the spatial decay lengths of the exchange interactions over a significantly denser parameter grid is necessary to fully map the exact phase boundaries. Nevertheless, evaluating the real-space Green's functions and executing the torque method integrations across a dense, multi-dimensional grid remains computationally costly.

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